

Components in wireless technology, 5XTC0

Module 4 Exercise: Amplifier design

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Where innovation starts

Outline

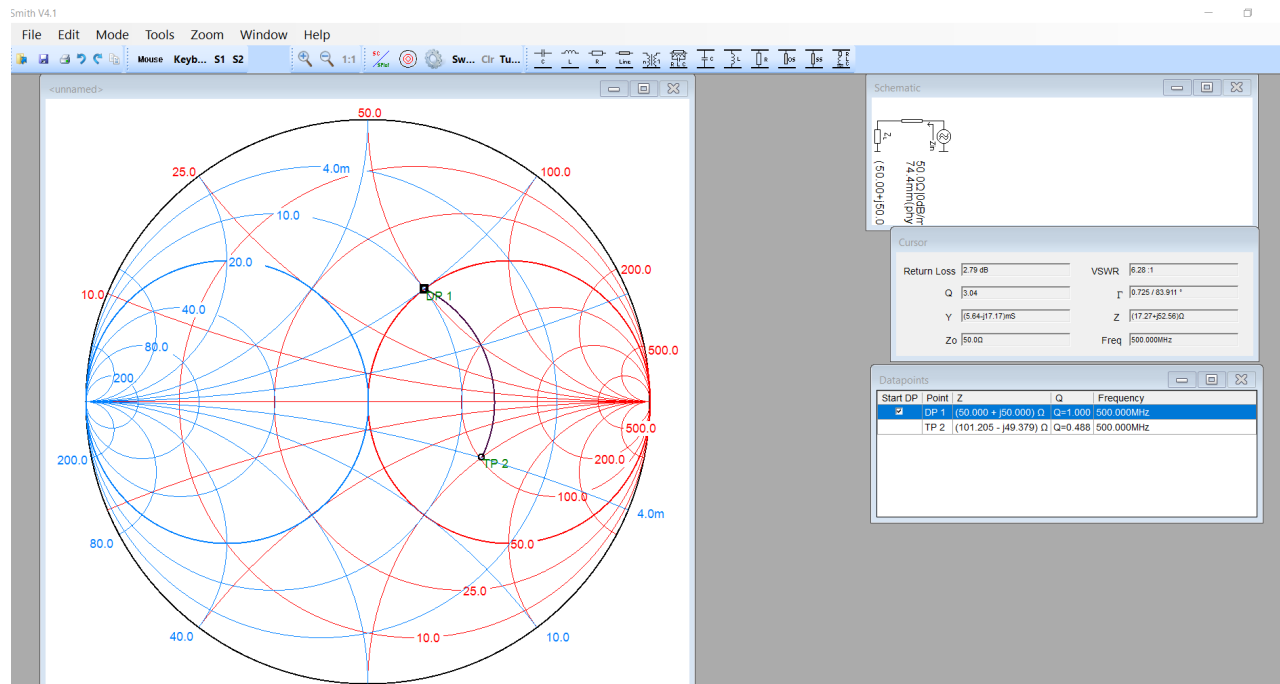
- Introduction to Smith Chart tool
- Introduction to formula list that can be used during exam
- Exercise:
 - Smith Chart
 - Transmission lines
 - Matching networks
 - Amplifier design

Smith Chart Tool

- Very good to practice and check analytical expressions
- Link to download the tool:

<https://www.fritz.dellsperger.net/smith.html>

- Smith 4.1

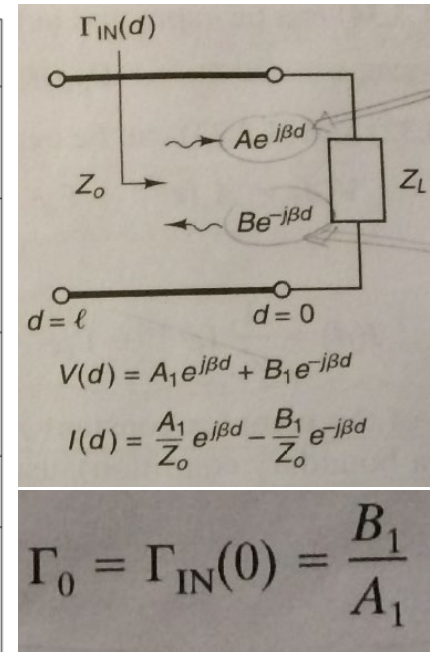


- Link to download Smith Chart: https://leleivre.com/rf_smith.html

Formula list available for exam

Power delivered to the load	$P_L = \frac{ V_2 ^2}{2Z_0} (1 - \Gamma_L ^2)$
Input power to the network	$P_{in} = \frac{ V_1 ^2}{2Z_0} (1 - \Gamma_{in} ^2)$
Input and output reflection coefficients of a transistor with a source and load: general case	$\Gamma_{in} = \frac{V_1^-}{V_1^+} = S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L}$ $\Gamma_{out} = \frac{V_2^-}{V_2^+} = S_{22} + \frac{S_{12}S_{21}\Gamma_s}{1 - S_{11}\Gamma_s}$
Input and output reflection coefficients of a transistor with a source and load: unilateral case	$\Gamma_{in} = \frac{V_1^-}{V_1^+} = S_{11}$ $\Gamma_{out} = \frac{V_2^-}{V_2^+} = S_{22}$
Gain of the input matching network	$G_S = \frac{1 - \Gamma_S ^2}{ 1 - \Gamma_{in}\Gamma_S ^2}$
Gain of the output matching network	$G_L = \frac{1 - \Gamma_L ^2}{ 1 - S_{22}\Gamma_L ^2}$
Gain of the transistor (unilateral case)	$G_0 = S_{21} ^2$
Transducer gain of the basic amplifier circuit (input matching, unilateral transistor, output matching)	$G_T = G_S G_0 G_L$ $G_{T,0dB} = G_{S,0dB} + G_{0,0dB} + G_{L,0dB}$
Maximum gain of the input and output matching networks	$G_{S,max} = \frac{1}{1 - S_{11} ^2}$ $G_{L,max} = \frac{1}{1 - S_{22} ^2}$
Maximum transducer power gain, unilateral case	$G_{TU,max} = \frac{1}{1 - S_{11} ^2} S_{21} ^2 \frac{1}{1 - S_{22} ^2}$
Normalized gain factors g_s and g_L	$g_s = \frac{G_S}{G_{S,max}} = \frac{1 - \Gamma_S ^2}{ 1 - S_{11}\Gamma_S ^2} (1 - S_{11} ^2)$ $g_L = \frac{G_L}{G_{L,max}} = \frac{1 - \Gamma_L ^2}{ 1 - S_{22}\Gamma_L ^2} (1 - S_{22} ^2)$
Center and radius of the constant gain circle for the input matching network	$C_S = \frac{g_S S_{11}^*}{1 - (1 - g_S) S_{11} ^2}$ $R_S = \frac{\sqrt{1 - g_S} (1 - S_{11} ^2)}{1 - (1 - g_S) S_{11} ^2}$
Center and radius of the constant gain circle for the output matching network	$C_L = \frac{g_L S_{22}^*}{1 - (1 - g_L) S_{22} ^2}$ $R_L = \frac{\sqrt{1 - g_L} (1 - S_{22} ^2)}{1 - (1 - g_L) S_{22} ^2}$
Condition for "unconditionally stable" device, general case	for all $ \Gamma_L < 1$ and $ \Gamma_S < 1$ $\Rightarrow \begin{cases} \Gamma_{in} = \left S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L} \right < 1 \\ \Gamma_{out} = \left S_{22} + \frac{S_{12}S_{21}\Gamma_S}{1 - S_{11}\Gamma_S} \right < 1 \end{cases}$

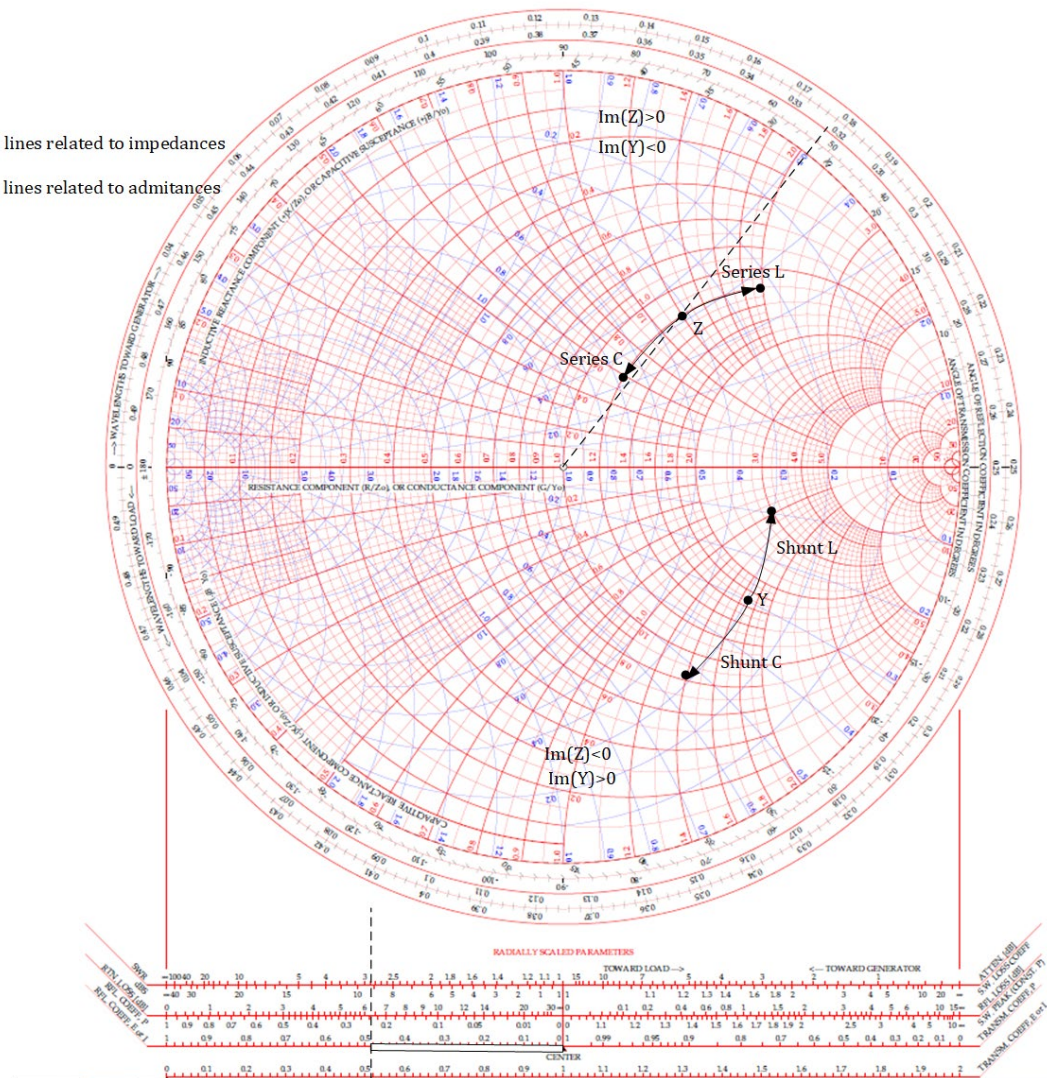
Conditions for "unconditionally stable" device, unilateral case	$ \Gamma_m = S_{11} < 1$ $ \Gamma_{out} = S_{22} < 1$
Center and radius of the stability circles, load side	$C_L = \frac{(S_{22} - \Delta S_{11}^*)^*}{ S_{22} ^2 - \Delta ^2}$ (center), $R_L = \frac{ S_{12}S_{21} }{ S_{22} ^2 - \Delta ^2}$ (radius), $\Delta = S_{11}S_{22} - S_{12}S_{21}$
Center and radius of the stability circles, source side	$C_S = \frac{(S_{11} - \Delta S_{22}^*)^*}{ S_{11} ^2 - \Delta ^2}$ (center), $R_S = \frac{ S_{12}S_{21} }{ S_{11} ^2 - \Delta ^2}$ (radius), $\Delta = S_{11}S_{22} - S_{12}S_{21}$
Test for unconditional stability, general case	$ \Delta = S_{11}S_{22} - S_{12}S_{21} < 1$ and $K = \frac{1 - S_{11} ^2 - S_{22} ^2 + \Delta ^2}{2 S_{12}S_{21} } > 1$
Test for unconditional stability, unilateral case	$ S_{11} < 1$ $ S_{22} < 1$
Two-stage amplifier: Output noise and noise figure	$P_{N,total} = G_{A2} (G_{A1} P_{N,m} + P_{n1}) + P_{n2}$ $\Rightarrow F_{total} = \frac{P_{N,total}}{P_{N,m} G_{A1} G_{A2}} = 1 + \frac{P_{n1}}{P_{N,m} G_{A1}} + \frac{P_{n2}}{P_{N,m} G_{A1} G_{A2}}$ $F_{total} = F_1 + \frac{F_2 - 1}{G_{A1}}$ with $F_j = 1 + \frac{P_{nj}}{P_{N,m} G_{Aj}}$, $j = 1, 2$
Noise figure of a 2-port amplifier	$F = F_{min} + \frac{r_N}{g_S} \left \frac{y_S - y_{opt}}{y_S} \right ^2$ $F = F_{min} + 4r_N \frac{ \Gamma_S - \Gamma_{opt} ^2}{(1 - \Gamma_S ^2) 1 + \Gamma_{opt} ^2}$
Constant noise circles	$C_F = \frac{\Gamma_{opt}}{1 + N}$ $R_F = \frac{1}{1 + N} \sqrt{N^2 + N(1 - \Gamma_{opt} ^2)}$ $\Delta F_n' = N = (F - F_{min}) \frac{ 1 + \Gamma_{opt} ^2}{4r_n} = \frac{ \Gamma_S - \Gamma_{opt} ^2}{1 - \Gamma_S ^2}$



Smith Chart Explanation

Red lines related to impedances

Blu lines related to admittances



Smith Chart: Problem 1

Problem 1: Locating impedances and admittances in the Smith chart, reading of reflection coefficients

- a) Find the following impedances in the Smith chart (normalized to $50\ \Omega$) and determine the corresponding reflection coefficient:

$Z_1 = 50\ \Omega$ (matched load)	$Z_2 = 0\ \Omega$ (short circuit)	$Z_3 = \infty$ (open circuit)
$Z_4 = 20\ \Omega + j\ 25\ \Omega$	$Z_5 = 20\ \Omega - j\ 30\ \Omega$	$Z_6 = 150\ \Omega - j\ 150\ \Omega$

- b) Find the following admittances in the Smith chart (normalized to $0.02\ \text{S}$) and determine the corresponding reflection coefficient:

$Y_1 = 0.02\ \text{S}$ (matched load)	$Y_2 = \infty$ (short circuit)	$Y_3 = 0$ (open circuit)
$Y_4 = 0.02 + j\ 0.02\ \text{S}$	$Y_5 = 0.05 - j\ 0.1\ \text{S}$	$Y_6 = 0.01 + j\ 0.05\ \text{S}$

Smith Chart: Solution for Problem 1a (1/2)

$Z_1 = 50 \Omega$ (matched load)	$Z_2 = 0 \Omega$ (short circuit)	$Z_3 = \infty$ (open circuit)
$Z_4 = 20\Omega + j 25\Omega$	$Z_5 = 20\Omega - j 30\Omega$	$Z_6 = 150\Omega - j 150\Omega$

Smith chart

Problem 1a

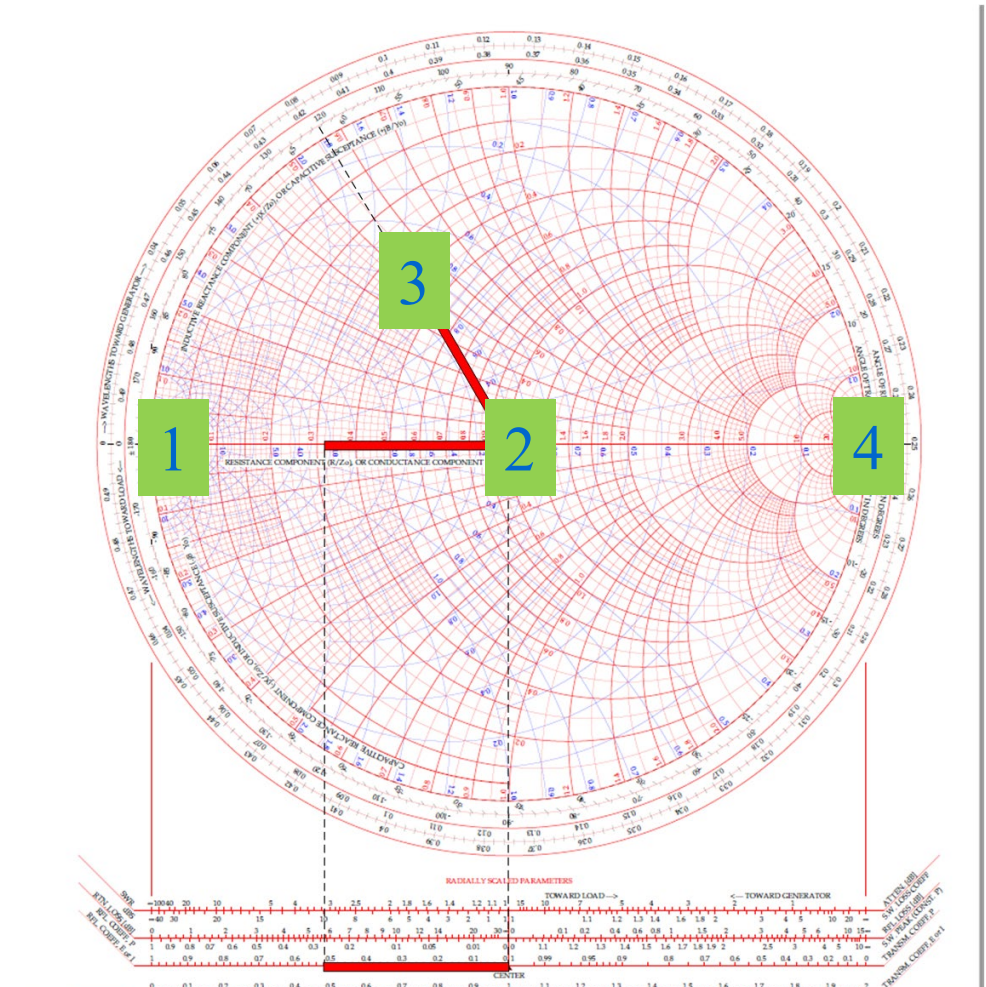
$Z_1 = 50$ $\Gamma_1 = \frac{Z_1 - Z_0}{Z_1 + Z_0} = 0$
 Smith: $\Gamma_{10} = \frac{50 - 50}{50} = 0$

$Z_2 = 0$ $\Gamma_2 = \frac{0 - Z_0}{0 + Z_0} = -1$
 Smith: $\Gamma_{20} = 0$

$Z_3 = \infty$ $\Gamma_3 = \frac{\infty - Z_0}{\infty} = 1$
 Smith: $\Gamma_{30} = \infty$

$Z_4 = 20 + j25$
 $Z_0 = 50$
 Smith: $Z_4 = 0.4 + j0.5$

$\Gamma_4 = \frac{20 + j25 - 50}{20 + j25 + 50}$
 $\Gamma_4 = \frac{-30 + j25}{70 + j25}$
 $\Gamma_4 = \frac{(-30 + j25)(70 - j25)}{70^2 + 25^2}$
 $\Gamma_4 = \frac{-2100 + j2050 - j1750 + 625}{4900 + 625}$
 $\Gamma_4 = \frac{-1475 + j2500}{5525}$
 $\Gamma_4 = 0.52 \angle 120.55^\circ$



Smith Chart: Solution for Problem 1a (2/2)

$Z_1 = 50 \Omega$ (matched load)	$Z_2 = 0 \Omega$ (short circuit)	$Z_3 = \infty$ (open circuit)
$Z_4 = 20 \Omega + j 25 \Omega$	$Z_5 = 20 \Omega - j 30 \Omega$	$Z_6 = 150 \Omega - j 150 \Omega$

Handwritten calculations for the reflection coefficient Γ_s for Z_5 and Z_6 .

For $Z_5 = 20 \Omega - j 30 \Omega$:

$$\Gamma_s = \frac{Z_5 - Z_0}{Z_5 + Z_0} = \frac{20 - j30 - 50}{20 - j30 + 50} = \frac{-30 - j30}{70 - j30}$$

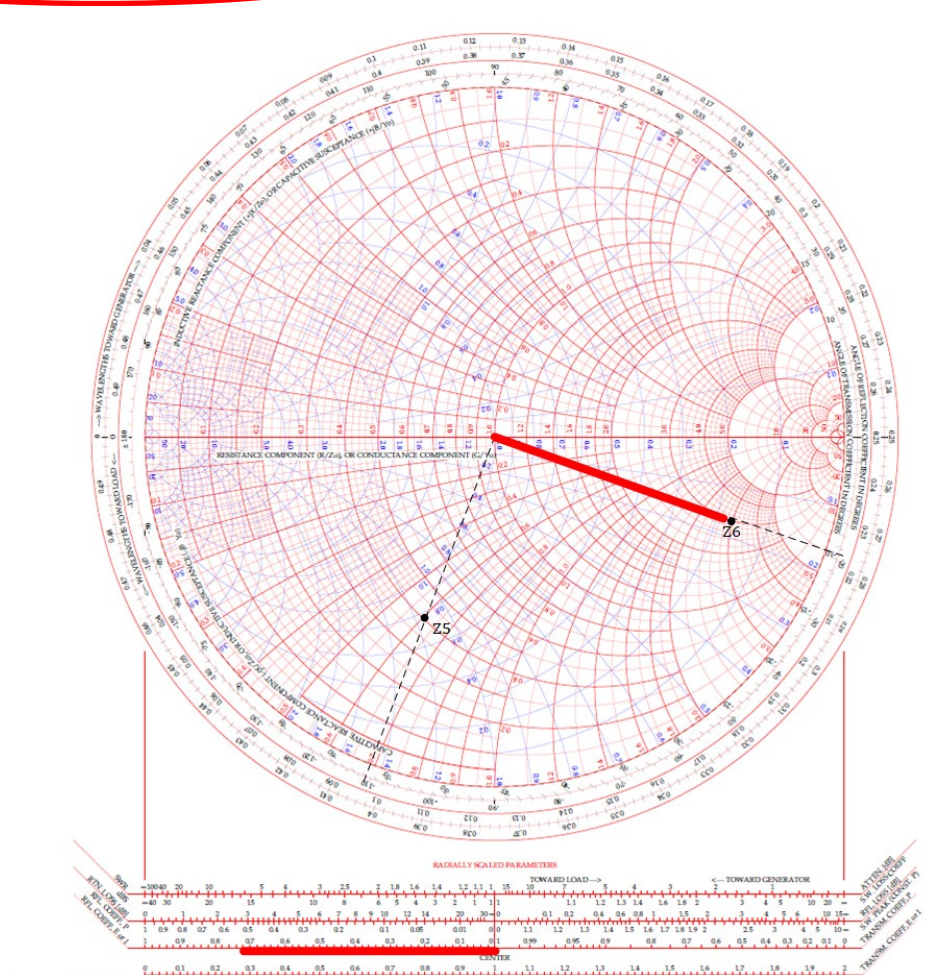
$$\Gamma_s = \frac{(-30 - j30)(70 + j30)}{(70 - j30)(70 + j30)} = \frac{-2100 - j2100}{4900 + 6300} = \frac{-2100 - j2100}{11200}$$

$$\Gamma_s = -0.1875 - j0.1875 = 0.265 \angle -135^\circ$$

For $Z_6 = 150 \Omega - j 150 \Omega$:

$$\Gamma_6 = \frac{Z_6 - Z_0}{Z_6 + Z_0} = \frac{150 - j150 - 50}{150 - j150 + 50} = \frac{100 - j150}{200 - j150}$$

$$\Gamma_6 = \frac{(100 - j150)(200 + j150)}{(200 - j150)(200 + j150)} = \frac{20000 - j15000 + j30000 + 22500}{40000 + 22500} = \frac{42500 - j15000}{62500}$$

$$\Gamma_6 = 0.68 - j0.24 = 0.72 \angle -19.1^\circ$$


Smith Chart: Solution for Problem 1b (1/3)

$$Y_1 = 0.02 \text{ S (matched load)}$$

$$Y_4 = 0.02 + j 0.02 \text{ S}$$

$$Y_2 = \infty \text{ (short circuit)}$$

$$Y_5 = 0.05 - j 0.1 \text{ S}$$

$$Y_3 = 0 \text{ (open circuit)}$$

$$Y_6 = 0.01 + j 0.05 \text{ S}$$

Smith Chart
Problem 1b

$$Z_{\text{Smith}} = \frac{Z}{Z_0} \quad Y_0 = 0.02 \text{ S}$$

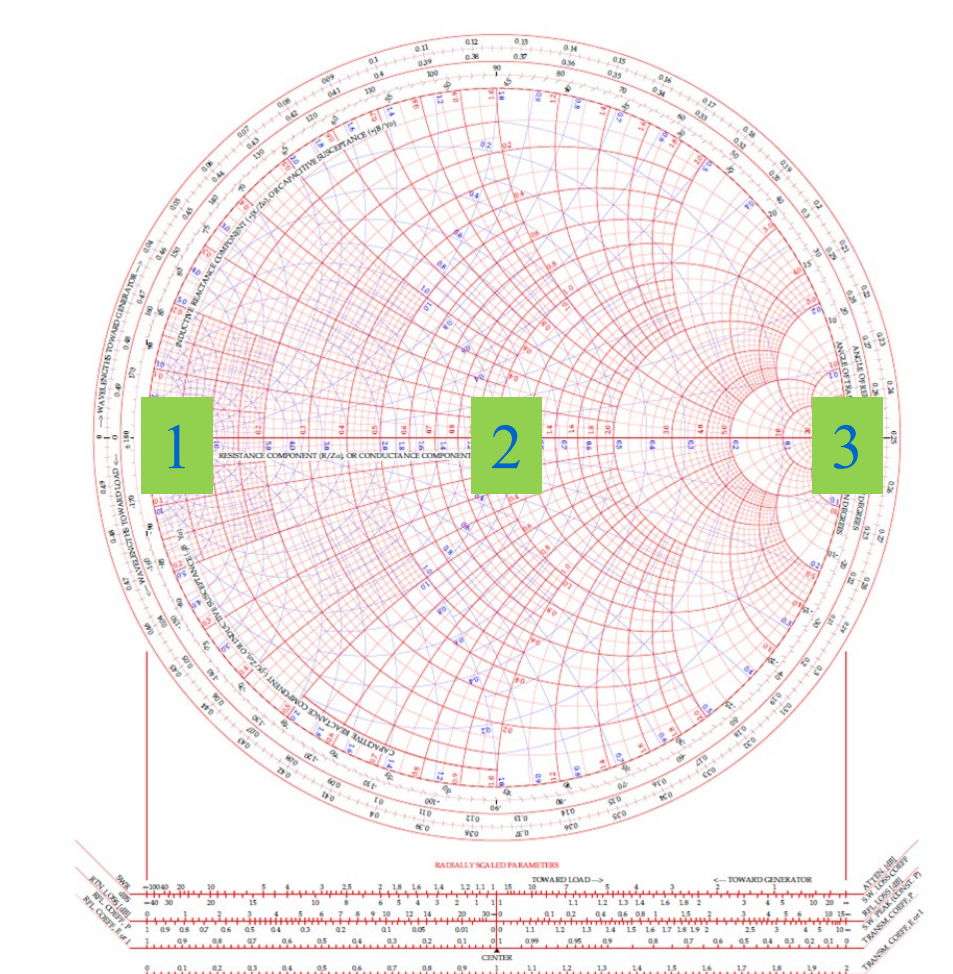
$$\frac{1}{Z_{\text{Smith}}} = \frac{Z_0}{Z} = \frac{1}{\frac{Z}{Z_0}} = \frac{Y}{Y_0}$$

$$\Gamma = \frac{Z - Z_0}{Z + Z_0} = \frac{\frac{Y}{Y_0} - 1}{\frac{Y}{Y_0} + 1} = \frac{Y - Y_0}{Y + Y_0}$$

$X_1 = 0.02 \text{ S}$ $\Gamma_1 = \frac{0.02 - 0.02}{0.02 + 0.02}$
 Smith $X_1 = \frac{0.02}{0.02}$ $\Gamma_1 = 0$
 $X_1 = 1$ $\Gamma_1 = 0$

$X_2 = \infty$ $\Gamma_2 = \frac{Y_0 - \infty}{Y_0 + \infty}$
 Smith $X_2 = \frac{\infty}{0.02}$ $\Gamma_2 = -1$
 $X_2 = \infty$ $\Gamma_2 = -1$

$X_3 = 0$ $\Gamma_3 = \frac{Y_0 - 0}{Y_0 + 0}$
 Smith $X_3 = 0$ $\Gamma_3 = 1$
 $X_3 = 0$ $\Gamma_3 = 1$



Smith Chart: Solution for Problem 1b (2/3)

$Y_1 = 0.02 \text{ S (matched load)}$

$Y_2 = \infty \text{ (short circuit)}$

$Y_3 = 0 \text{ (open circuit)}$

$Y_4 = 0.02 + j 0.02 \text{ S}$

$Y_5 = 0.05 - j 0.1 \text{ S}$

$Y_6 = 0.01 + j 0.05 \text{ S}$

$$X_4 = 0.02 + j 0.02$$

$$\text{Smith } X_4 = \frac{0.02 + j 0.02}{0.02}$$

$$X_4 = 1 + j 1$$

$$\Gamma_4 = \frac{y_0 - y}{y_0 + y}$$

$$\Gamma_4 = \frac{0.02 - 0.02 - j 0.02}{0.02 + 0.02 + j 0.02}$$

$$\Gamma_4 = \frac{-j 0.02}{0.04 + j 0.02}$$

$$\Gamma_4 = \frac{-j}{2 + j}$$

$$\Gamma_4 = \frac{-j(2 - j)}{2^2 + 1^2}$$

$$\Gamma_4 = \frac{-1 - j 2}{5} = -0.2 - j 0.4$$

$$X_5 = 0.05 - j 0.1$$

$$\text{Smith } X_5 = \frac{0.05 - j 0.1}{0.02}$$

$$X_5 = 2.5 - j 5$$

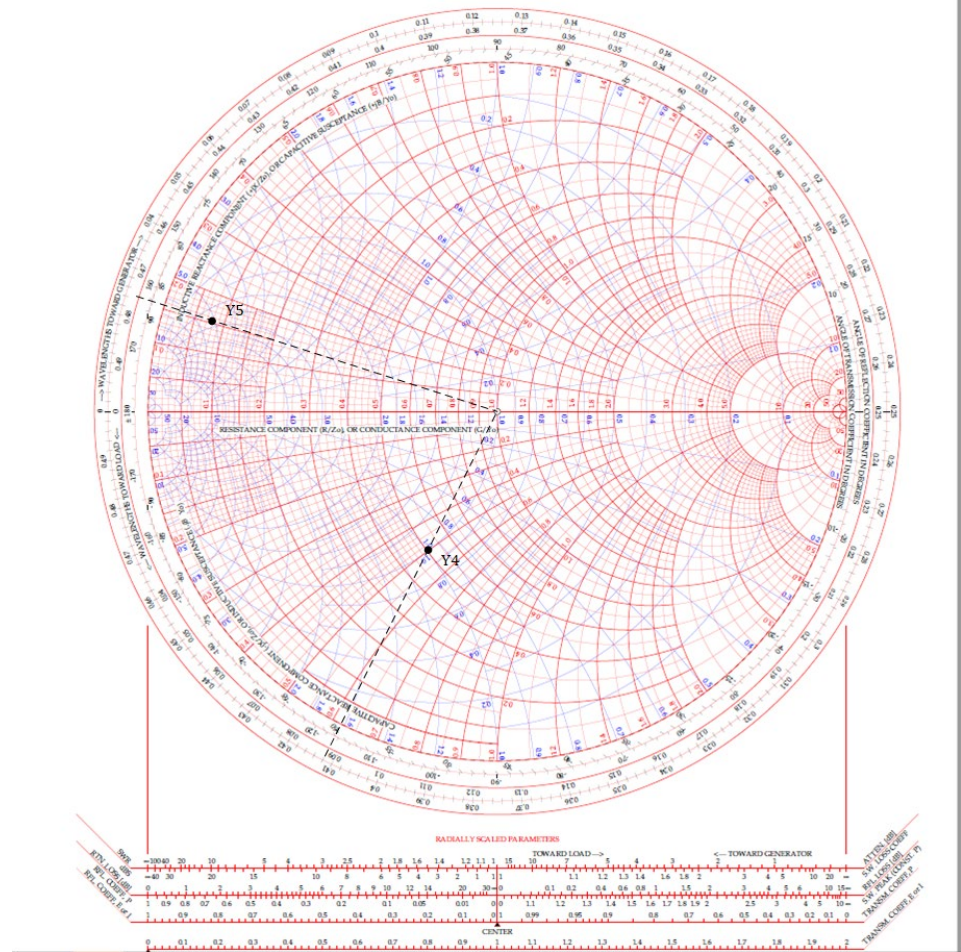
$$\Gamma_5 = \frac{y_0 - y}{y_0 + y}$$

$$\Gamma_5 = \frac{0.02 - 0.05 + j 0.1}{0.02 + 0.05 - j 0.1}$$

$$\Gamma_5 = \frac{-0.03 + j 0.1}{0.07 - j 0.1}$$

$$\Gamma_5 = \frac{-12.5 + j 40}{149}$$

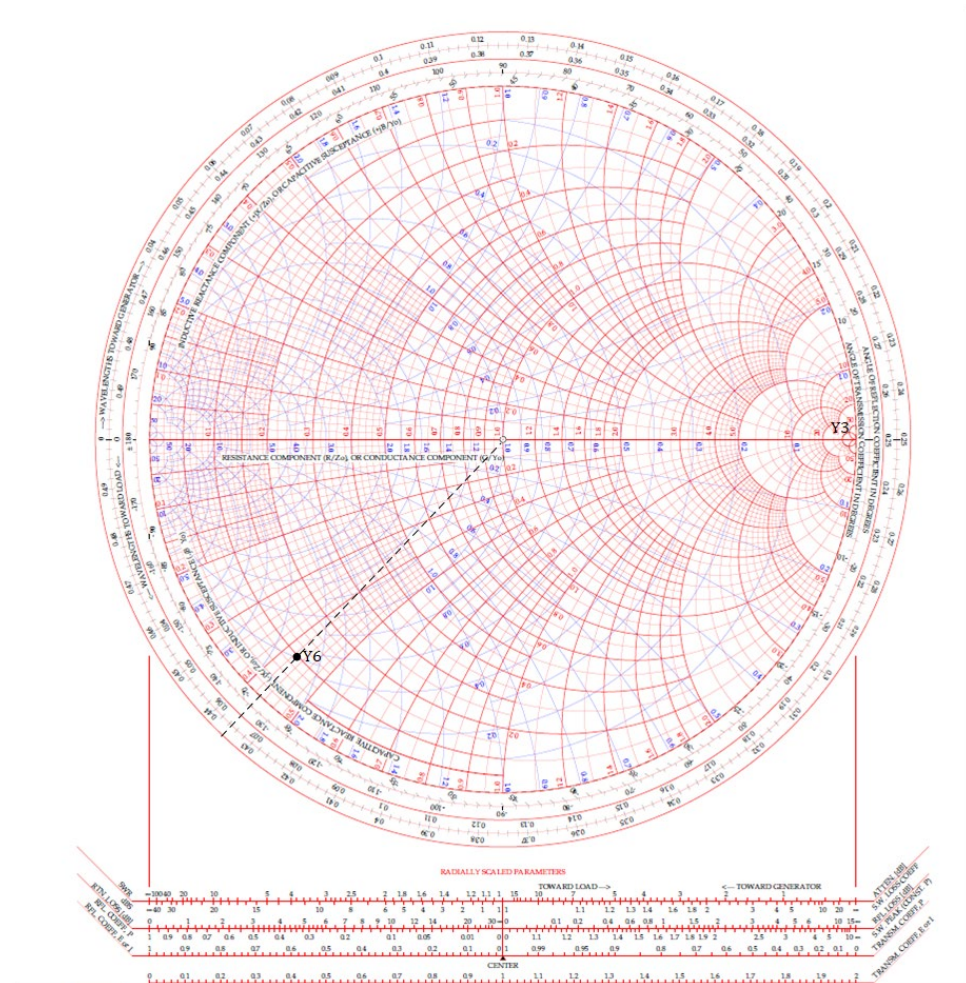
$$\Gamma_5 = 0.95 \angle 161.5^\circ$$



Smith Chart: Solution for Problem 1b (3/3)

$Y_1 = 0.02 \text{ S (matched load)}$	$Y_2 = \infty \text{ (short circuit)}$	$Y_3 = 0 \text{ (open circuit)}$
$Y_4 = 0.02 + j 0.02 \text{ S}$	$Y_5 = 0.05 - j 0.1 \text{ S}$	$Y_6 = 0.01 + j 0.05 \text{ S}$

$Y_6 = 0.01 + j 0.05$
 $Y_6 = \frac{0.01 + j 0.05}{0.02}$
 $Y_6 = 0.5 + j 2.5$
 $\Gamma_6 = \frac{0.02 - 0.01 - j 0.05}{0.02 + 0.01 + j 0.05}$
 $\Gamma_6 = \frac{0.01 - j 0.05}{0.03 + j 0.05} \cdot 100$
 $\Gamma_6 = \frac{1 - j 5}{3 + j 5} \cdot \frac{3 - j 5}{3 - j 5}$
 $\Gamma_6 = \frac{-24 - j 20}{39}$
 $\Gamma_6 = 0.87 e^{j 134.7^\circ}$



Smith Chart: Problem 2

Problem 2: Transforming impedances into admittances

- a) Find the corresponding normalized admittance values for the normalized impedance of $z = 1 + j1$.
- b) Analytically calculate the normalized admittance value and compare it to your reading of the Smith chart from question a).

Smith Chart: Solution for Problem 2

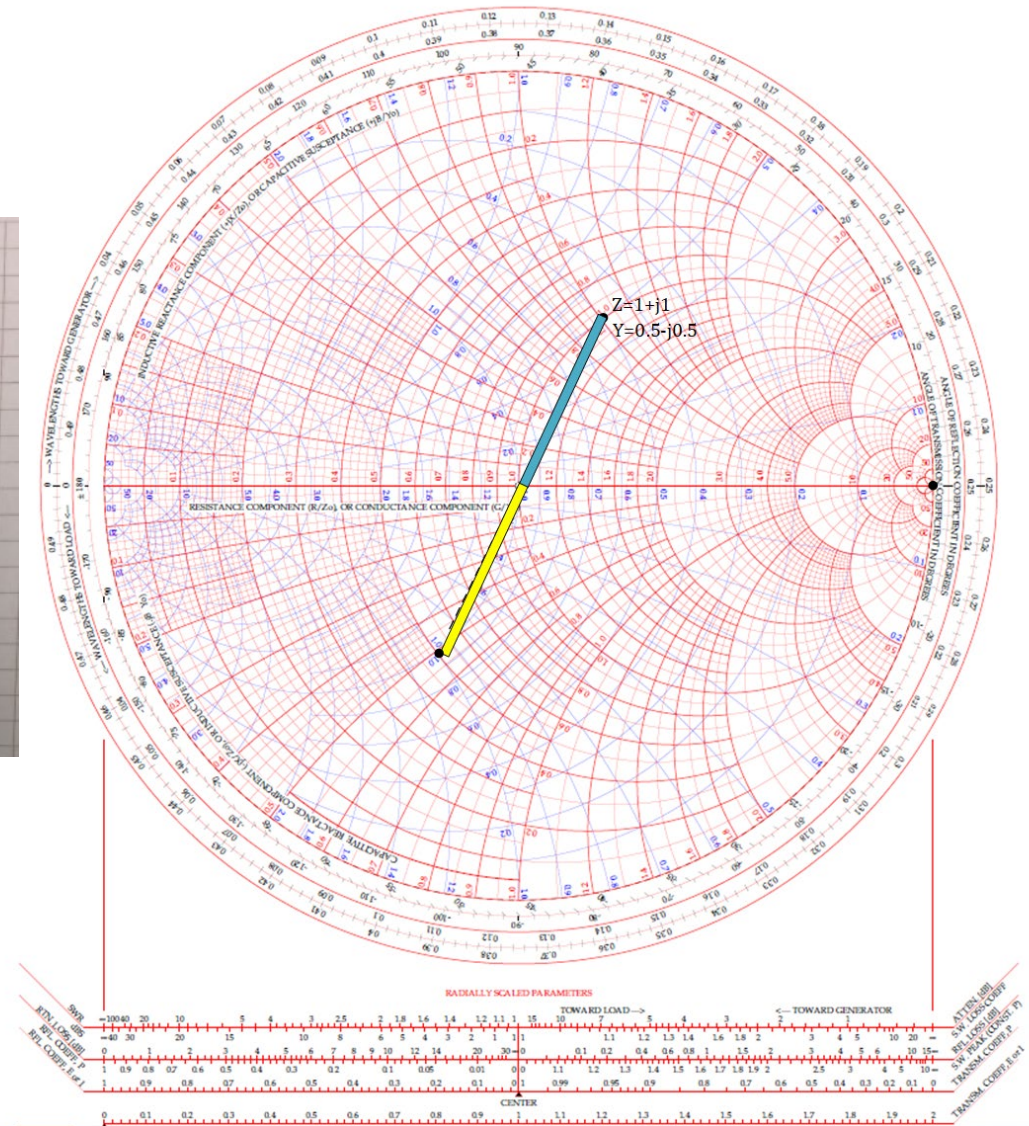
Smith Chart: Problem 2

$$Z = 1 + j1$$

$$Y = \frac{1}{Z}$$

$$Y = \frac{1}{1 + j1} = \frac{1 - j1}{(1 - j1)(1 + j1)} = \frac{1 - j1}{2}$$

$$Y = 0.5 - j0.5$$



Smith Chart: Problem 3

Problem 3: From reflection coefficient to impedance and VSWR

Find the normalized input impedance whose reflection coefficient is $0.5 \angle 26^\circ$.

Determine the corresponding VSWR for this reflection coefficient in the Smith chart and by analytical calculation. Hint: Use the scale (SWR) in the Smith chart.

Smith Chart: Solution for Problem 3

Smith Chart: Problem 3

$$\Gamma = 0.5 \angle 26^\circ$$

$$V_{SWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

$$V_{SWR} = \frac{1 + 0.5}{1 - 0.5} = 3$$

$Z = 2.2 + j1.2 \rightarrow$ Reading from Smith Chart $\rightarrow$ normalized impedance

$$\Gamma = \frac{Z - Z_0}{Z + Z_0}$$

$$\Gamma \cdot Z + \Gamma Z_0 = Z - Z_0$$

$$Z(1 - \Gamma) = Z_0(1 + \Gamma)$$

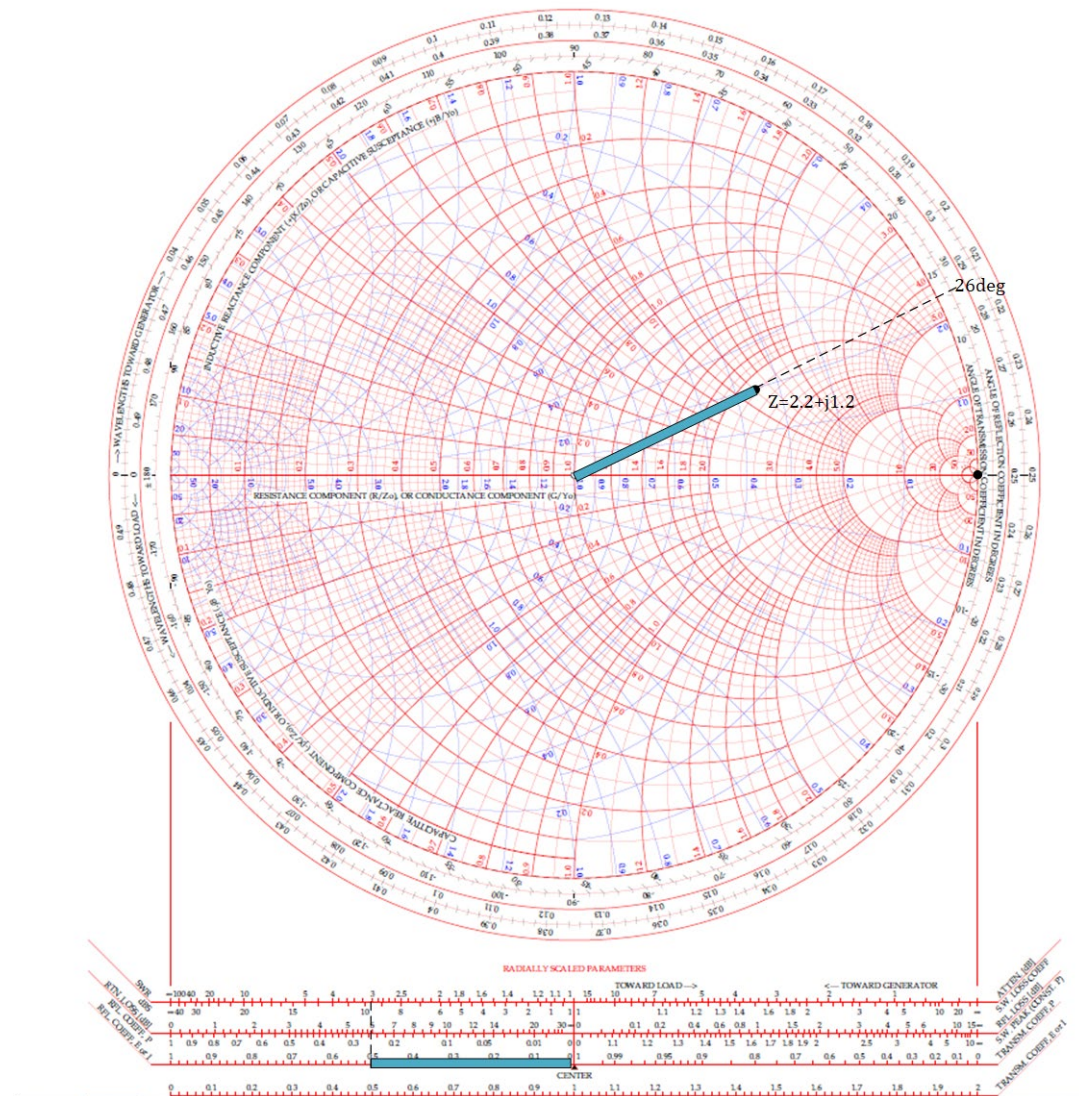
$$Z = Z_0 \frac{1 + \Gamma}{1 - \Gamma}$$

$$Z = Z_0 \cdot \frac{1 + 0.5 \angle 26^\circ}{1 - 0.5 \angle 26^\circ}$$

$$\frac{Z}{Z_0} = \frac{1 + 0.5(\cos 26^\circ + j \sin 26^\circ)}{1 - 0.5(\cos 26^\circ + j \sin 26^\circ)}$$

$$\frac{Z}{Z_0} = \frac{1 + 0.45 + j0.22}{1 - 0.45 - j0.22}$$

$$\frac{Z}{Z_0} = \frac{(1.45 + j0.22)(0.55 + j0.22)}{0.55^2 + 0.22^2}$$



$$\frac{Z}{Z_0} = \frac{0.75 + j0.44}{0.35}$$

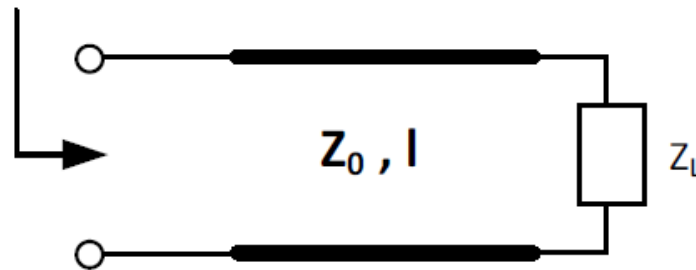
$$\frac{Z}{Z_0} = 2.14 + j1.25$$

Transmission lines: Problem 4

- a) Use the Smith chart to find the input reflection coefficient, the input impedance and the VSWR for a transmission line having an electrical length of 45° and characteristic impedance of $50\ \Omega$ and terminated in a load of $Z_L = 50 + j50\ \Omega$.
- b) Compare the graphical solution of a) to an analytical calculation of the values.

Γ : input reflection coefficient

Z_{IN} : input impedance



Transmission lines: Solution for Problem 4 (1/2)

Smith chart - Problem 4

a) $\beta l = \frac{\pi}{4} \rightarrow$ electrical length
 $\beta d = \pi/4$

$\frac{2\pi}{\lambda} l = \frac{\pi}{4}$
 $l = \frac{\lambda}{8} = 0.125 \lambda$

$Z_L = 50 + j50$

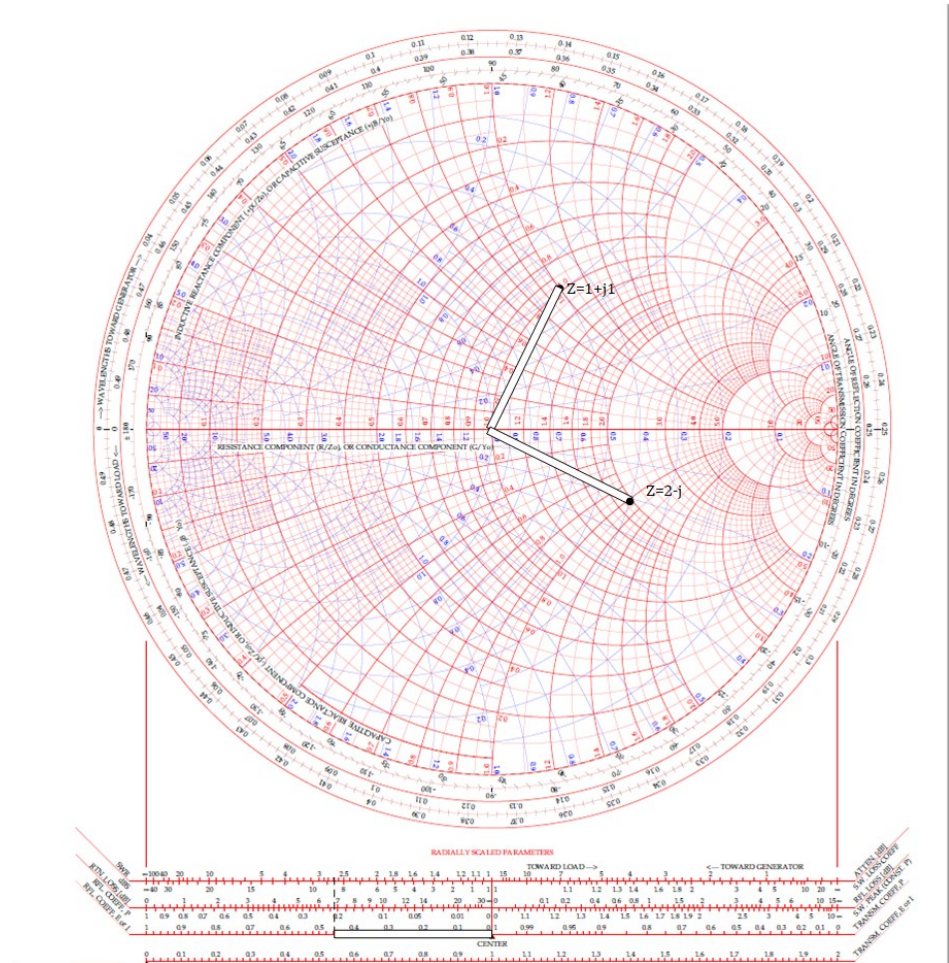
$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan \beta d}{Z_0 + jZ_L \tan \beta d} \quad / \quad d=l$

$Z_{in} = Z_0 \frac{50 + j50 + j50}{50 + j(50 + j50)}$

$Z_{in} = Z_0 \frac{50 + j100}{j50}$

$Z_{in} = Z_0 \frac{1 + j2}{j} \quad / \quad \cdot -j$

$Z_{in} = Z_0 (2 - j)$



Transmission lines: Standing wave Ratio (SWR)

$$T_o = \frac{Z_L - Z_o}{Z_L + Z_o}$$

$$T_o = \frac{50 + j50 - 50}{50 + j50 + 50}$$

$$T_o = \frac{j50}{100 + j50} = \frac{j}{2+j}$$

$$T_o = \frac{1+j2}{5}$$

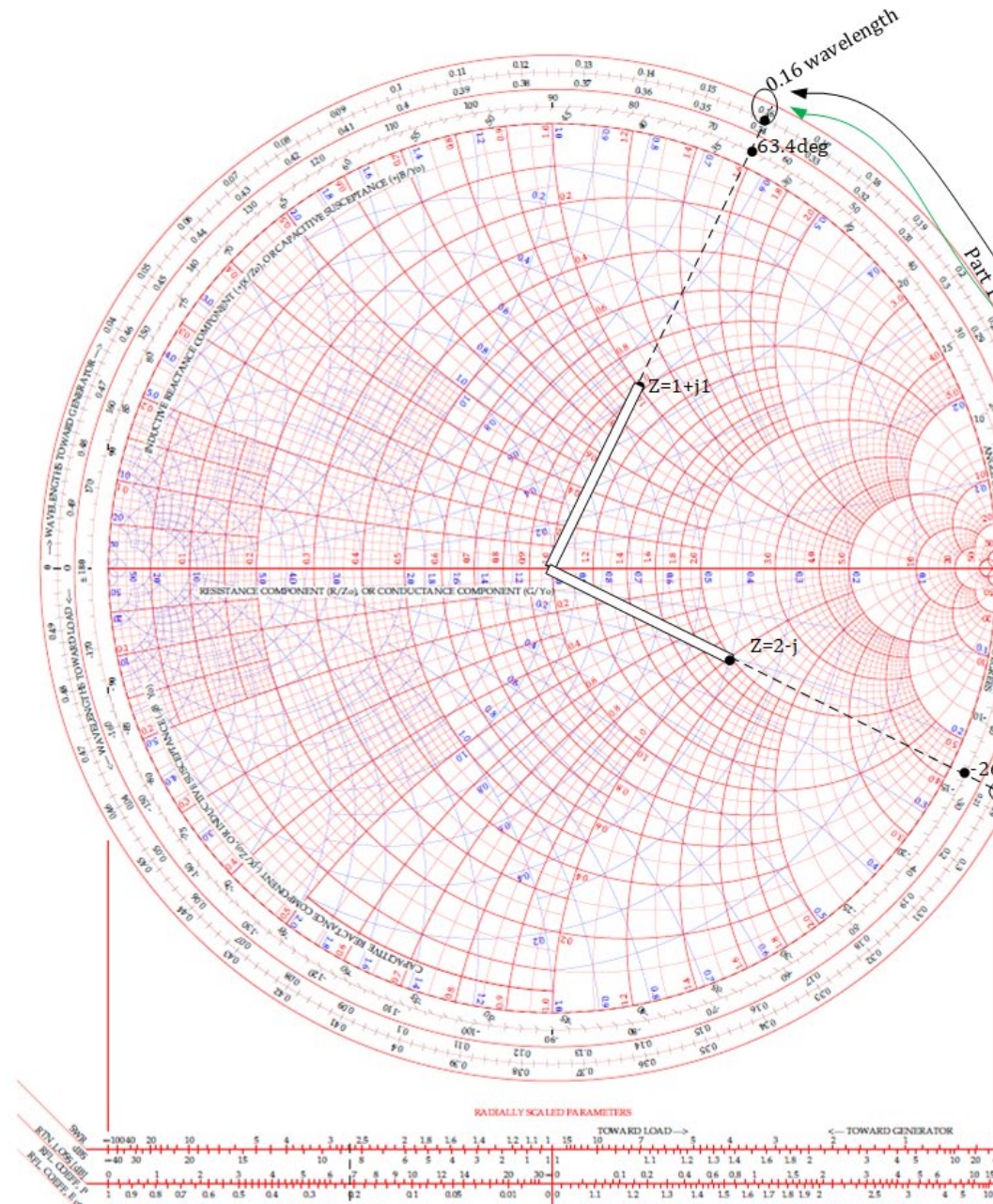
$$T_o = 0.44 e^{j63.43}$$

$$V_{SWR} = \frac{1 + 0.44}{1 - 0.44} = 2.57$$

$$T_w = T_o e^{-j2\beta d} = T_o e^{-j\pi/2}$$

$$T_w = 0.44 e^{j(63 - \pi/2)}$$

$$T_w = 0.44 e^{j26.57}$$



Transmission lines: Problem 5

- 5) Analytically calculate the required length l of a $50\ \Omega$ short circuited transmission line to obtain an input impedance of $Z_{in}(l) = j\ 100\ \Omega$. Express the length in terms of the wavelength λ . Determine the required length in the Smith chart and compare to your calculation.

Transmission lines: Solution for Problem 5

$$Z_{in} = Z_0 \left(\frac{Z_L + j Z_0 \tan \beta d}{Z_0 + j Z_L \tan \beta d} \right)^{-1}$$

$$Z_L = 0$$

$$Z_{in} = Z_0 \left(\frac{Z_0}{j Z_0 \tan \beta d} \right)^{-1}$$

$$Z_{in} = \cancel{j Z_0 \cot \beta d}$$

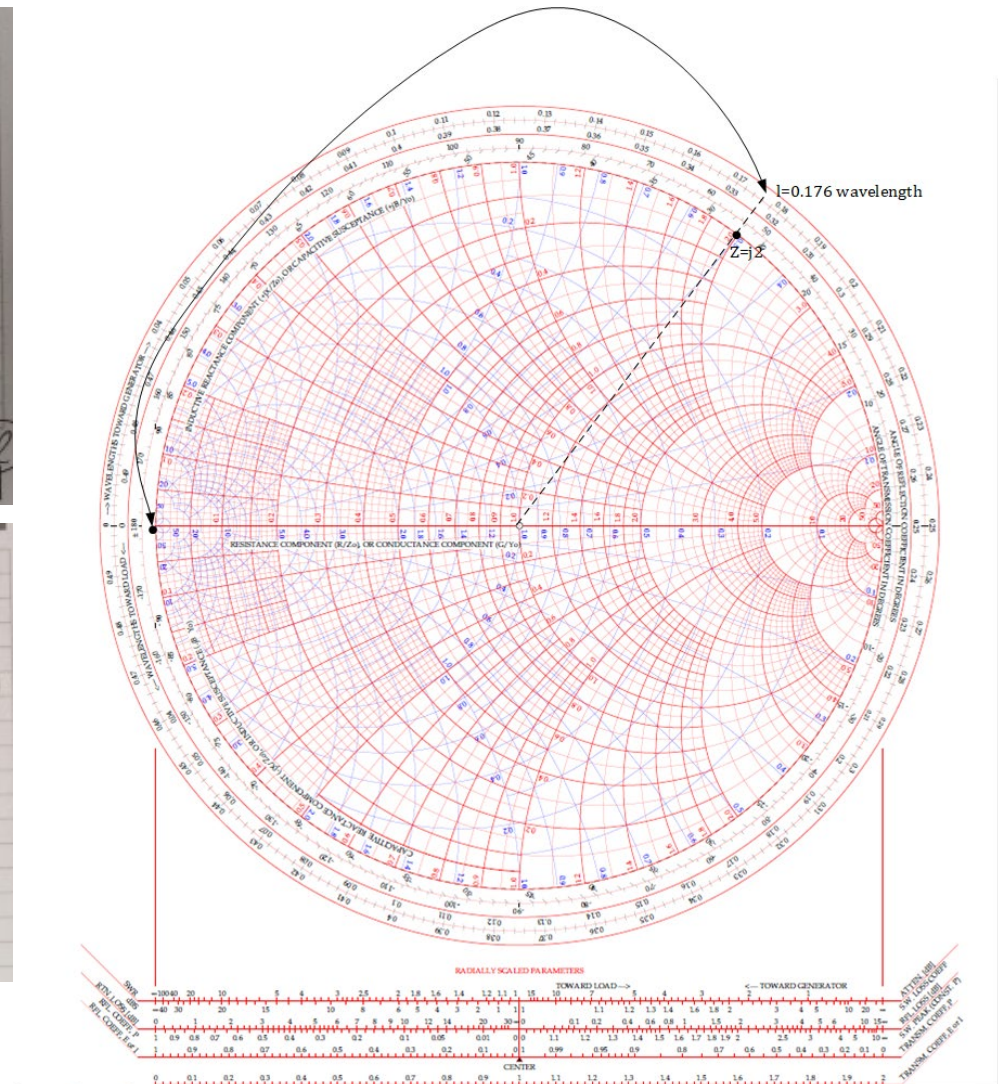
$$Z_{in} = j Z_0 \tan \beta d$$

$$j 100 = j 50 \tan \frac{2\pi \cdot d}{\lambda}$$

$$2 = \tan \frac{2\pi \cdot d}{\lambda} \quad (\pi = 180^\circ)$$

$$\tan^{-1} 2 = 63.43^\circ = 2 \cdot 180^\circ \cdot \frac{d}{\lambda}$$

$$d = 0.176 \lambda$$



Transmission lines: Problem 6

- 6) Analytically calculate the required length l of a $50\ \Omega$ open circuited transmission line to obtain an input impedance of $Z_{in}(l) = j\ 100\ \Omega$. Express the length in terms of the wavelength λ . Determine the required length in the Smith chart and compare to your calculation.

Transmission lines: Solution for Problem 6

$$d) \quad Z_{in} = Z_0 \frac{Z_L + j Z_0 \tan \beta d}{Z_0 + j Z_L \tan \beta d}$$

$$Z_L = \infty$$

$$Z_{in} = Z_0 \frac{1}{j \tan \beta d}$$

$$Z_{in} = Z_0 (-j \cot \beta d)$$

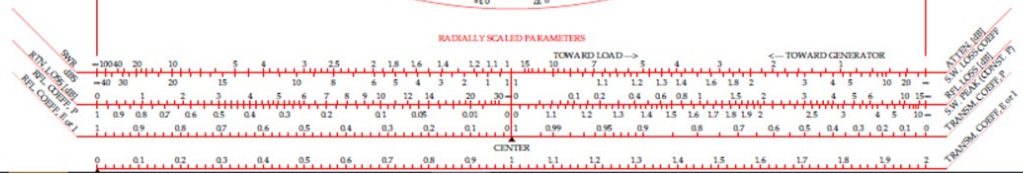
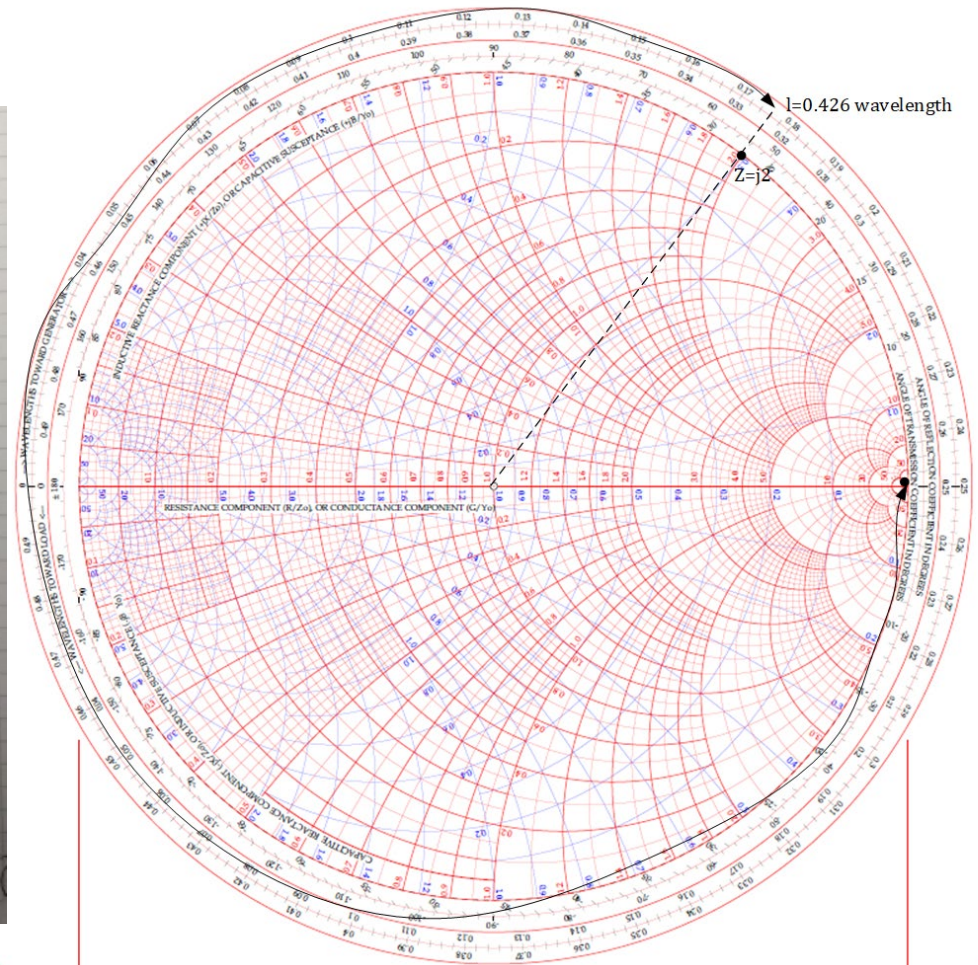
$$j_{100} = -50 j \cot \beta d$$

$$-2 = \cot \beta d$$

$$\beta d = \cot^{-1}(-2)$$

$$2\pi \cdot d = 153.44$$

$$d = 0.426 \lambda$$



Transmission lines: Problem 7

- Book of Gonzalez: Example 1.3.1

- (a) Find the load reflection coefficient, the input impedance, and the VSWR in the transmission line shown in Fig. 1. The length of the transmission line is $\lambda/8$ and the characteristic impedance is $50\ \Omega$.
- (b) Evaluate $V(\lambda/8)$, $I(\lambda/8)$, $P(\lambda/8)$, $V(0)$, $I(0)$ and $P(0)$.
- (c) Write the expression for $v(d,t)$ and $i(d,t)$ along the transmission line.
- (d) Find the length in centimeters of the $\lambda/8$ transmission line at $f=1\text{GHz}$.

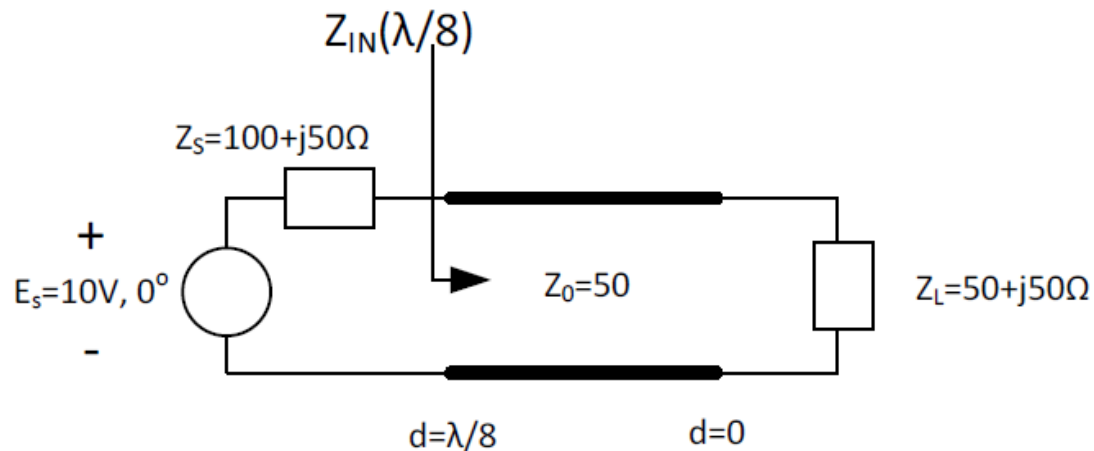
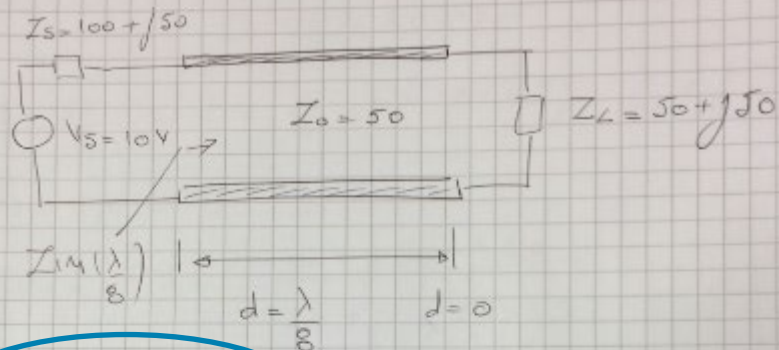


Figure 1 Transmission line circuit example problem 1.

Transmission lines: Solution for Problem 7 a

Gonzales: Example 1.3.1



a)

$$\Gamma_0 = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$\Gamma_0 = \frac{50 + j50 - 50}{50 + j50 + 50} = \frac{j50}{100 + j50} = \frac{j}{2 + j}$$

$$\Gamma_0 = \frac{j(2 - j)}{(2 + j)(2 - j)} = \frac{1 + j2}{5}$$

$$|\Gamma_0| = \frac{\sqrt{1 + 2^2}}{5} = \frac{1}{\sqrt{5}} = 0.447$$

$$\angle \Gamma_0 = \tan^{-1}(2/1) = 63.43$$

$$\Gamma_0 = 0.447 \angle 63.43$$

1

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan \beta d}{Z_0 + jZ_L \tan \beta d}$$

$$\beta = \frac{2\pi}{\lambda}, d = \frac{\lambda}{8} \Rightarrow \beta d = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{8} = \frac{\pi}{4}$$

$$\tan \pi/4 = 1$$

$$Z_{in} = Z_0 \frac{50 + j50 + j50}{50 + j(50 + j50)}$$

$$Z_{in} = Z_0 \frac{50 + j100}{j50}$$

$$Z_{in} = 8 \cdot 100 + 50$$

$$Z_{in} = 100 - j50$$

$$V = A e^{j\beta d} + B e^{-j\beta d}$$

$$V = A e^{j\beta d} (1 - \Gamma_0 e^{-j2\beta d})$$

$$|V_{max}| = |A| (1 + |\Gamma_0|)$$

$$|V_{min}| = |A| (1 - |\Gamma_0|)$$

$$VSWR = \frac{|V_{max}|}{|V_{min}|} = \frac{1 + |\Gamma_0|}{1 - |\Gamma_0|}$$

$$VSWR = \frac{1 + 0.447}{1 - 0.447} = 2.62$$

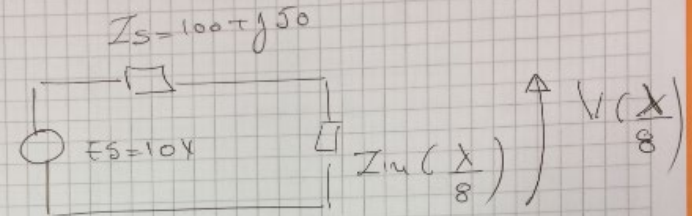
2

Transmission lines: Solution for Problem 7 b (1/3)

$$\begin{aligned}
 b) \quad V &= A_1 e^{j\beta d} + B_1 e^{-j\beta d} \\
 V &= A_1 e^{j\beta d} \left(1 + \frac{B_1}{A_1} e^{-j2\beta d} \right) \\
 T_0 &= \frac{B_1}{A_1} \\
 V(d) &= A_1 e^{j\beta d} (1 + T_0 e^{-j2\beta d}) \\
 I(d) &= \frac{A_1}{Z_0} e^{-j\beta d} (1 - T_0 e^{-j2\beta d}) \\
 d &= \lambda/8 \quad \beta d = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{8} = \pi/4 \\
 V\left(\frac{\lambda}{8}\right) &= A_1 e^{j\pi/4} (1 + 0.447 e^{j63.43^\circ} e^{-j\pi/2}) \\
 V\left(\frac{\lambda}{8}\right) &= A_1 e^{j\pi/4} (1 + 0.447 e^{-j26.57^\circ}) \\
 V\left(\frac{\lambda}{8}\right) &= A_1 (e^{j\pi/4} + 0.447 e^{j18.43^\circ}) \\
 V\left(\frac{\lambda}{8}\right) &= A_1 (0.71 + j0.71 + 0.42 + j0.14) \\
 V\left(\frac{\lambda}{8}\right) &= A_1 (1.13 + j0.85)
 \end{aligned}$$

3

Equivalent circuit:



$$Z_{in}\left(\frac{\lambda}{8}\right) = 100 - j50$$

$$V\left(\frac{\lambda}{8}\right) = E_s \cdot \frac{Z_{in}\left(\frac{\lambda}{8}\right)}{Z_s + Z_{in}\left(\frac{\lambda}{8}\right)}$$

$$V\left(\frac{\lambda}{8}\right) = E_s \cdot \frac{100 - j50}{100 + j50 + 100 - j50}$$

$$V\left(\frac{\lambda}{8}\right) = E_s \cdot \frac{100 - j50}{200}$$

$$V\left(\frac{\lambda}{8}\right) = 10 (0.5 - j0.25)$$

$$V\left(\frac{\lambda}{8}\right) = 10 \cdot \sqrt{0.5^2 + 0.25^2} \angle \tan^{-1} \frac{-0.25}{0.5}$$

$$V\left(\frac{\lambda}{8}\right) = 5.59 \angle -26.57^\circ$$

4

Transmission lines: Solution for Problem 7 b (2/3)

$$I(\frac{\lambda}{8}) = \frac{E_s}{Z_s + Z_{in}(\frac{\lambda}{8})}$$

$$I(\frac{\lambda}{8}) = \frac{10}{100 + j50 + 100 - j50}$$

$$I(\frac{\lambda}{8}) = \frac{10}{200} = 0.05 \text{ A}$$

$$P(\frac{\lambda}{8}) = \operatorname{Re} \left(V_{\text{rms}}(\frac{\lambda}{8}) \cdot I_{\text{rms}}(\frac{\lambda}{8})^* \right)$$

$$P(\frac{\lambda}{8}) = \frac{1}{2} \operatorname{Re} \left(V(\frac{\lambda}{8}) \cdot I(\frac{\lambda}{8})^* \right)$$

$$P(\frac{\lambda}{8}) = \frac{1}{2} \operatorname{Re} (10 \cdot (0.5 - j0.25) \cdot 0.05)$$

$$P(\frac{\lambda}{8}) = \frac{1}{2} 10 \cdot 0.5 \cdot 0.05$$

$$P(\frac{\lambda}{8}) = 125 \text{ mW}$$

$$V(\frac{\lambda}{8}) = 10(0.5 - j0.25) \rightarrow \text{calculated on page 9}$$

$$V(\frac{\lambda}{8}) = A_1 (1.13 + j0.85) \rightarrow \text{calculated on page 3}$$

$$A_1 = ?$$

$$10(0.5 - j0.25) = A_1 (1.13 + j0.85)$$

$$A_1 = 10 \frac{0.5 - j0.25}{1.13 + j0.85}$$

$$A_1 = 10 \frac{(0.5 - j0.25)(1.13 - j0.85)}{(1.13)^2 + (0.85)^2}$$

$$A_1 = 10 \frac{0.3525 - j0.4075}{2}$$

$$A_1 = 3.95 \angle -63.45^\circ$$

$$V(d) = 3.95 \angle -63.45^\circ e^{j\beta d} (1 + 0.447 \angle 63.43^\circ e^{-j2\beta d})$$

$$V(d) = 3.95 \angle -63.45^\circ (1 + 0.447 \angle 63.43^\circ e^{-j2\beta d})$$

$$V(d) = 3.95 \angle -63.45^\circ e^{j\beta d} + 1.77 e^{-j\beta d}$$

$$I(d) = \frac{3.95 \angle -63.45^\circ e^{j\beta d} - 1.77 e^{-j\beta d}}{50}$$

(6)

(5)

Transmission lines: Solution for Problem 7 b (3/3)

$$\begin{aligned}
 V(d) &= 3.95 e^{-j63.45} + 1.77 e^{j63.45} \\
 V(0) &\Rightarrow d=0 \\
 V(0) &= 3.95 e^{-j63.45} + 1.77 e^{j63.45} \\
 V(0) &= 3.95 \cos(63.45) - j 3.95 \sin(63.45) + 1.77 \cos(63.45) + j 1.77 \sin(63.45) \\
 V(0) &= 3.53 - j 3.53 \\
 \boxed{V(0) = 3.53 (1 - j) = 5 \cdot e^{-j45^\circ}} \\
 I(0) &= \frac{V(0)}{50 + j50} = \frac{3.53(1-j)}{50(1+j)}
 \end{aligned}$$

$$\begin{aligned}
 P(0) &= \frac{1}{2} \operatorname{Re} \{ V(0) \cdot I(0)^* \} \\
 P(0) &= \frac{1}{2} \operatorname{Re} \{ 5 e^{-j45^\circ} \cdot 0.0706 e^{j45^\circ} \} \\
 P(0) &= \frac{1}{2} \cdot 0.353 \cos(45^\circ) \\
 P(0) &= 125 \text{ mW}
 \end{aligned}$$

Transmission lines: Solution for Problem 7 c,d

$$c) V(d,t) = \operatorname{Re} \left\{ (3.55 e^{j\beta d} - 63.45 e^{j\beta d} + 1.77 e^{-j\beta d}) \cdot e^{j\omega t} \right\}$$

$$V(d,t) = \operatorname{Re} \left\{ 3.55 e^{j\beta d - j63.45 \omega t} + 1.77 e^{-j\beta d + j\omega t} \right\}$$

$$|V(d,t)| = 3.55 \cos(\omega t + \beta d - 63.45) + 1.77 \cos(\omega t - \beta d)$$

$$V(d,t) = 3.55 [\cos(\omega t + \beta d) \cos(63.45) + \sin(\omega t + \beta d) \sin(63.45)] + 1.77 \cos(\omega t - \beta d)$$

$$V(d,t) = 2.55 \cos(\omega t + \beta d) + 3.53 \sin(\omega t + \beta d) + 1.77 \cos(\omega t - \beta d)$$

$$V(d,t) = 1.76 (\cos \omega t \cos \beta d - \sin \omega t \sin \beta d) + 3.53 (\sin \omega t \cos \beta d + \cos \omega t \sin \beta d) + 1.77 (\cos \omega t \cos \beta d + \sin \omega t \sin \beta d)$$

$$V(d,t) = 3.54 \cos \beta d \cos \omega t + 3.54 \sin \beta d \cos \omega t + 3.54 \cos \beta d \sin \omega t \quad (7)$$

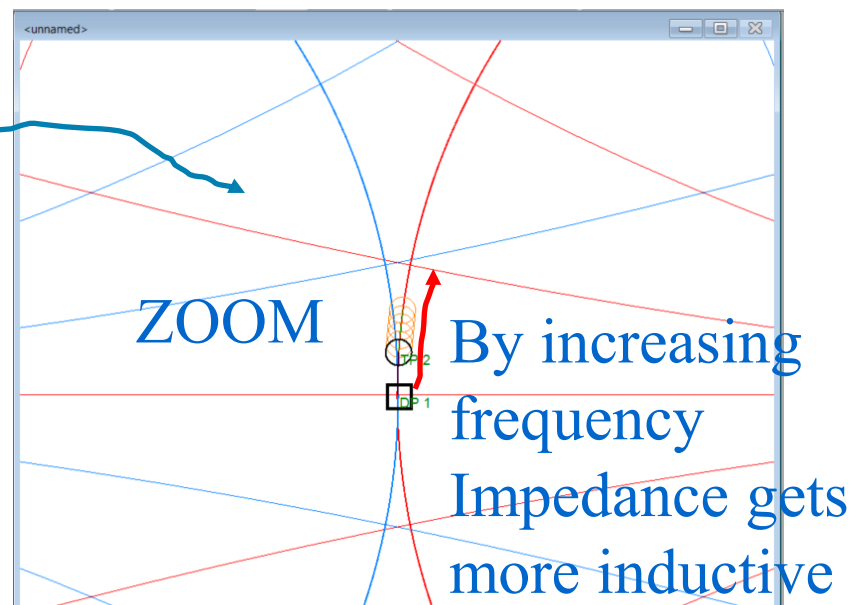
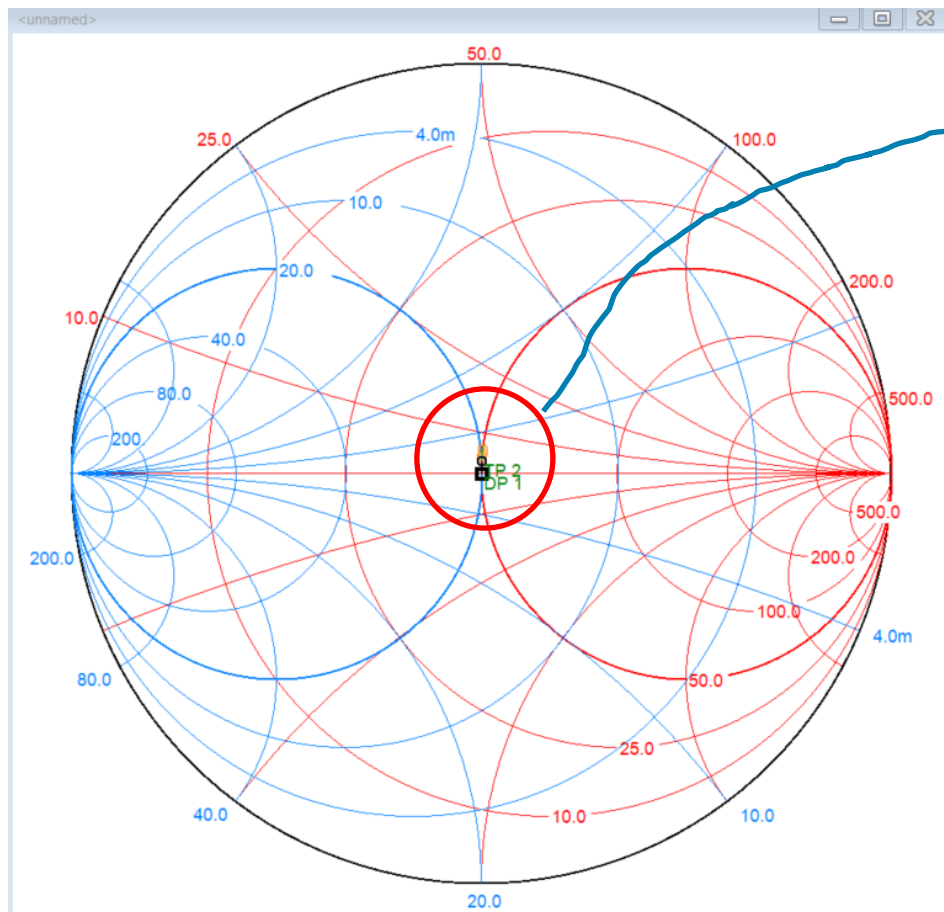
$$d) f = 1 \text{ GHz}$$

$$\lambda = c/f = 0.3$$

$$\frac{\lambda}{8} = 3.75 \text{ cm.}$$

Matching network: Problem 9

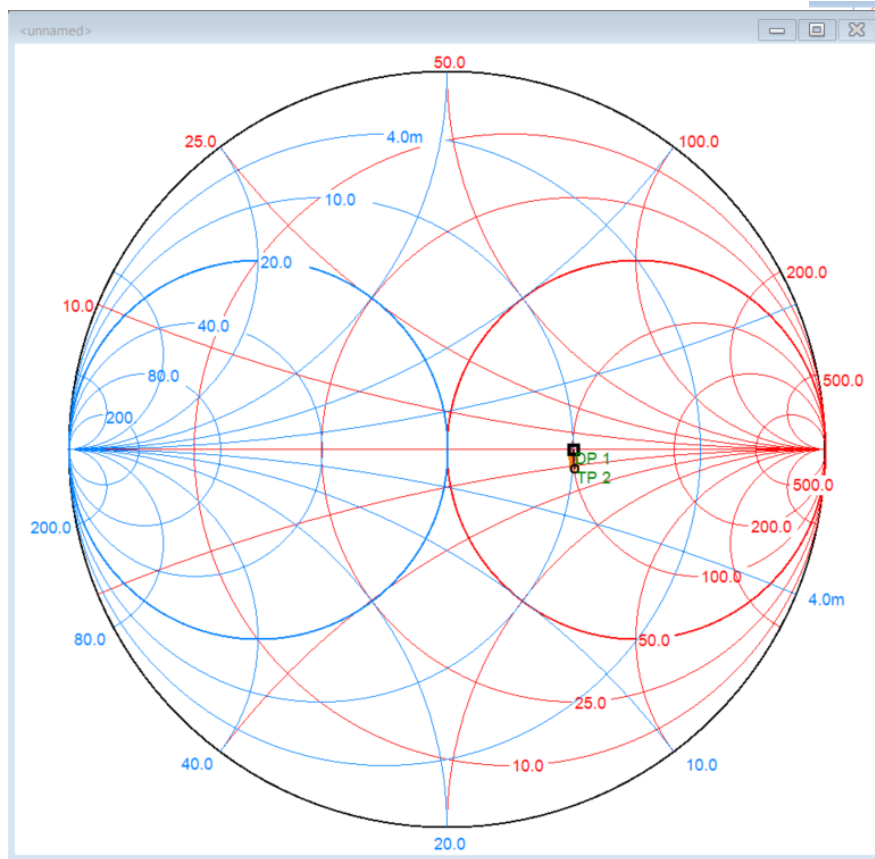
For a series R-L network draw the path in the ZY-Smith chart for the frequency range 500 MHz to 1 GHz for $R=50\ \Omega$ and $L=1\ \text{nH}$.



Start DP	Point	Z	Q	Frequency
	TP 2	(50.000 + j3.250) Ω	Q=0.065	500.000MHz
	SP 1	(50.000 + j3.250) Ω	Q=0.065	500.000MHz
	SP 2	(50.000 + j3.899) Ω	Q=0.078	600.000MHz
	SP 3	(50.000 + j4.549) Ω	Q=0.091	700.000MHz
	SP 4	(50.000 + j5.199) Ω	Q=0.104	800.000MHz
	SP 5	(50.000 + j5.849) Ω	Q=0.117	900.000MHz
	SP 6	(50.000 + j6.499) Ω	Q=0.130	1.000GHz

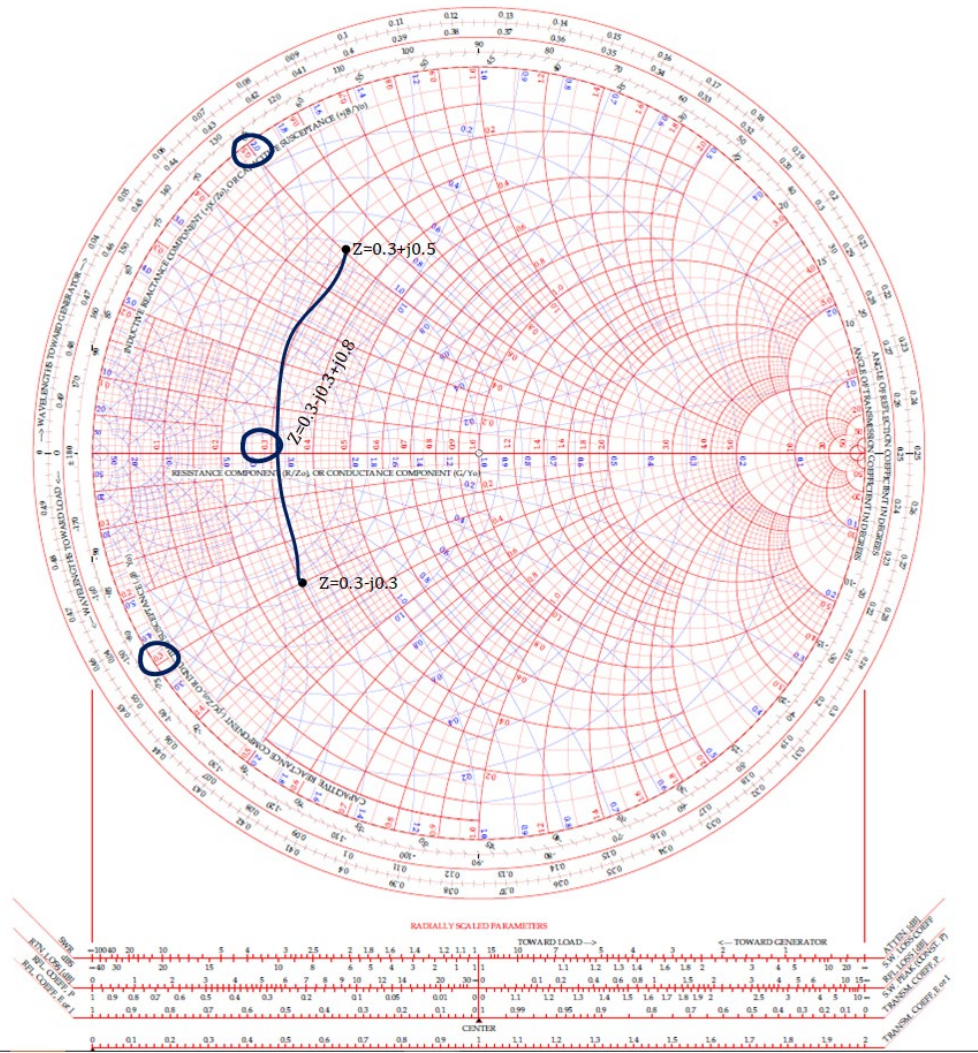
Matching network: Problem 10

For a series R-C network draw the path in the 50 Ω ZY-Smith chart for the frequency range 900 MHz to 2.5 GHz for $R=100\ \Omega$ and $C=1.5\ \text{pF}$.



Matching: Problem 11

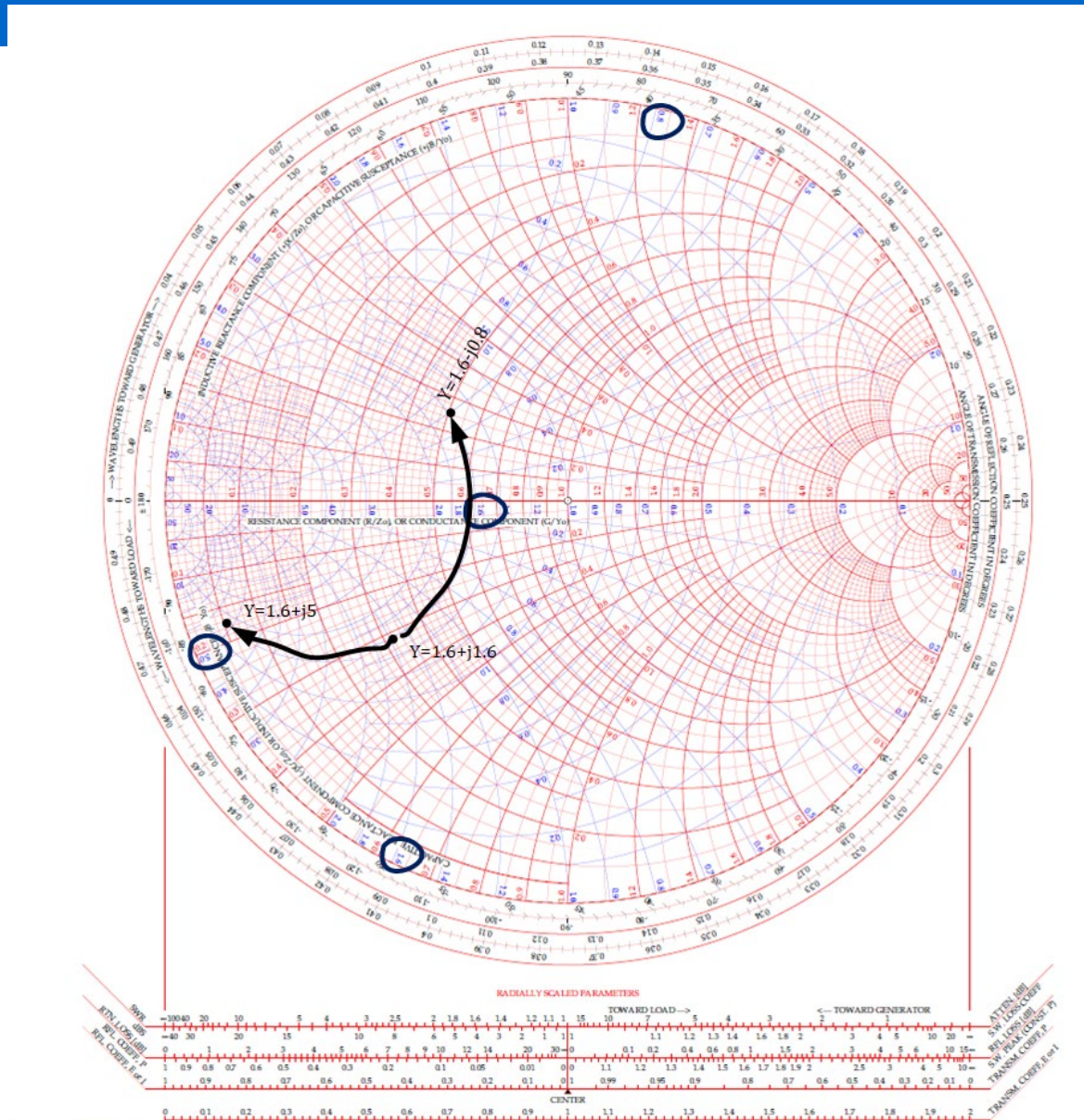
In the Smith chart, illustrate the effect of adding a series inductor L ($z_L = j0.8$) to an impedance $z = 0.3 - j0.3$.



Matching: Problem 12

- d) Illustrate the effect of adding a shunt inductor L ($y_L = -j2.4$) to an admittance $y = 1.6 + j1.6$ in the ZY Smith chart.
- e) Illustrate the effect of adding a shunt capacitor C ($y_{LC} = j3.4$) to an admittance $y = 1.6 + j1.6$ in the ZY Smith chart.

Matching: Problem 12 - solution

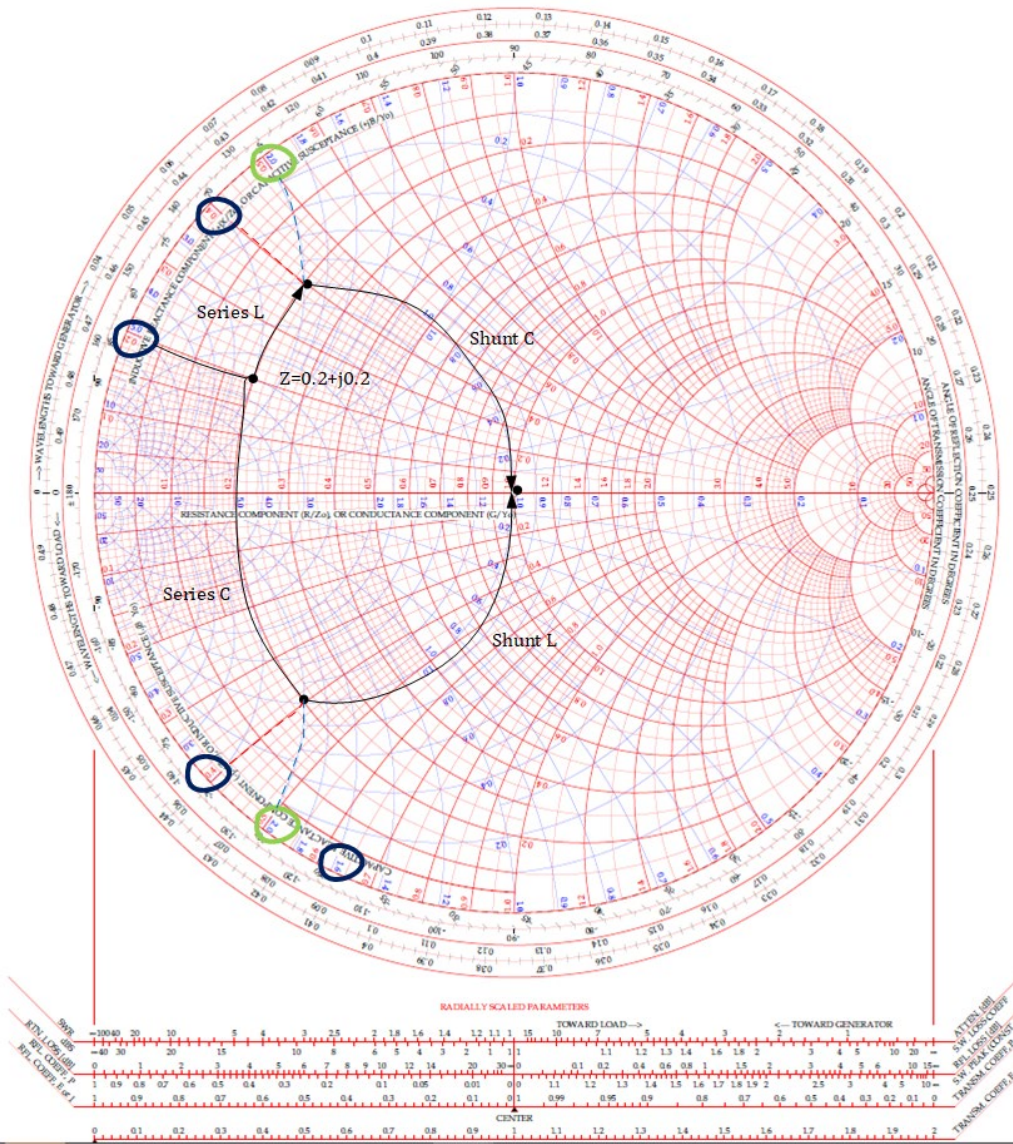


Smith Chart: Problem 13

A load of impedance $Z_{\text{LOAD}} = 10 + j10\Omega$ is to be matched to a 50Ω line for the frequency $f=500$ MHz.

- a) Use the Smith chart and design two 2-element L-C matching networks and specify the values of L and C. Illustrate your answer in the Smith chart.
- b) Find the values of L and C by an analytical calculation (without the Smith chart).

Matching network: Problem 13 – solution (1/2)



a) option 1:

$$Z_{in} = 50 \Omega$$

$$Z_L = 10 \Omega + j10 \Omega$$

$$Z = 0.2 + j0.2$$

$$\omega L = 0.4 - 0.2 = 0.2 \cdot 50 \Omega$$

$$L = \frac{10 \Omega}{2\pi \cdot 500 \cdot 10^6 \text{ Hz}} = 3.18 \text{ nH}$$

$$\omega C = 2 \cdot 0.02 = 0.04$$

$$C = \frac{0.04}{2\pi \cdot 500 \cdot 10^6} = 12.7 \text{ pF}$$

option 2:

$$Z_{in} = 50 \Omega$$

$$Z_L = 10 \Omega + j10 \Omega$$

$$Z = 0.2 + j0.2$$

$$\frac{1}{\omega C} = (0.2 - (-0.4)) \cdot 50 \Omega = 30 \Omega$$

$$\rightarrow C = \frac{1}{2\pi \cdot 500 \cdot 10^6 \cdot 30 \Omega} = 10.6 \text{ pF}$$

$$\frac{1}{\omega L} = 2 \cdot 0.02 \text{ S} = 0.04 \text{ S}$$

$$L = \frac{1}{2\pi \cdot 500 \cdot 10^6 \cdot 0.04} = 7.96 \text{ nH}$$

Matching network: Problem 13 – solution (2/2)

b) option 1:

$$50 \Omega = Z_{in} = ((Z_L + j\omega L)^{-1} + j\omega C)^{-1}$$

$$0,02 = \frac{1}{Z_L + j\omega L} + j\omega C$$

$$= \frac{1}{R + j\text{Im}(Z_L) + j\omega L} + j\omega C$$

$$= \frac{R - j(\text{Im} Z_L + \omega L)}{R^2 + (\text{Im} Z_L + \omega L)^2} + j\omega C$$

$$\text{condition 1: } \frac{R}{R^2 + (\text{Im} Z_L + \omega L)^2} = 0,02 \quad (1)$$

$$\text{condition 2: } \frac{-\text{Im} Z_L - \omega L}{R^2 + (\text{Im} Z_L + \omega L)^2} + \omega C = 0 \quad (2)$$

$$(1) \Rightarrow R = 0,02 (R^2 + (\text{Im} Z_L + \omega L)^2)$$

$$\pm \sqrt{\frac{R}{0,02} - R^2} = (\text{Im} Z_L + \omega L)$$

$$\Rightarrow \omega L = \pm \sqrt{\frac{R}{0,02} - R^2} - \text{Im} Z_L$$

$$\omega L = (\pm 20 - 10) \Omega$$

$$= 10 \Omega, \text{ as } \omega L > 0$$

$$L = 3,18 \cdot 10^{-9} \text{ H}$$

$$= 3,18 \text{ nH}$$

$$(2) \text{ with } L = 3,18 \text{ nH: } \omega C = \frac{\text{Im} Z_L + \omega L}{R^2 + (\text{Im} Z_L + \omega L)^2} = 0,04$$

$$C = 12,7 \text{ pF}$$

Similar procedure for option ②

Matching network: Problem 14

- a) Design a matching network for $f = 5$ GHz composed of an inductor L and a capacitor C to transform a $50\ \Omega$ load at the input to an admittance $Y_{\text{OUT}} = (8 - j12) \cdot 10^{-3}\ \text{S}$ at the output.
- b) Illustrate your solution in the Smith chart.

Matching network: Problem 14 –solution (1/2)

$f=5\text{GHz}$

$$2\pi f C / 0.02 = 0.6$$

$$C = 0.6 * 0.02 / (2\pi f)$$

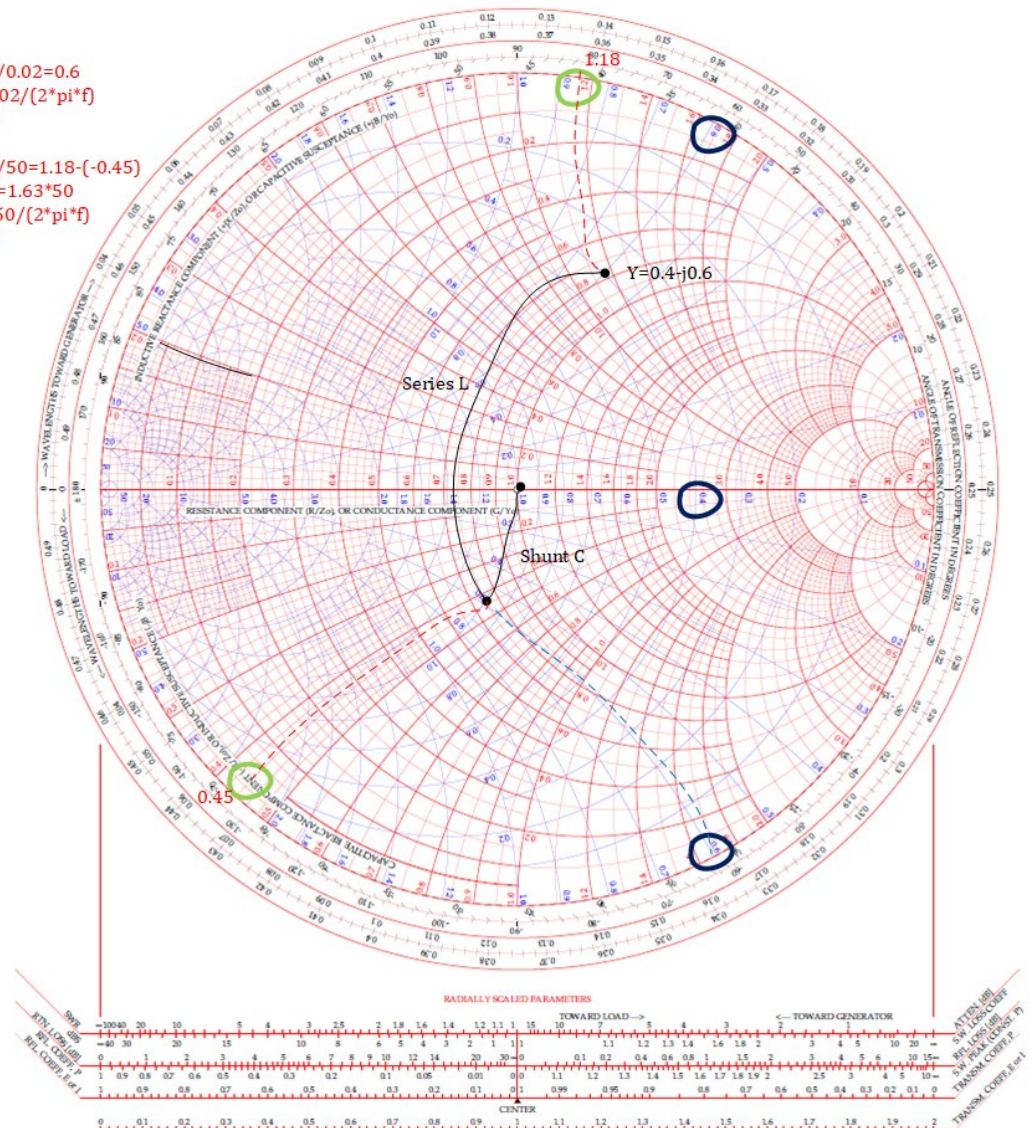
$$C = 382\text{fF}$$

$$2\pi f L / 50 = 1.18 \rightarrow (-0.45)$$

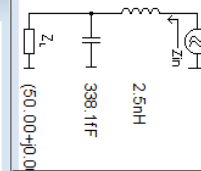
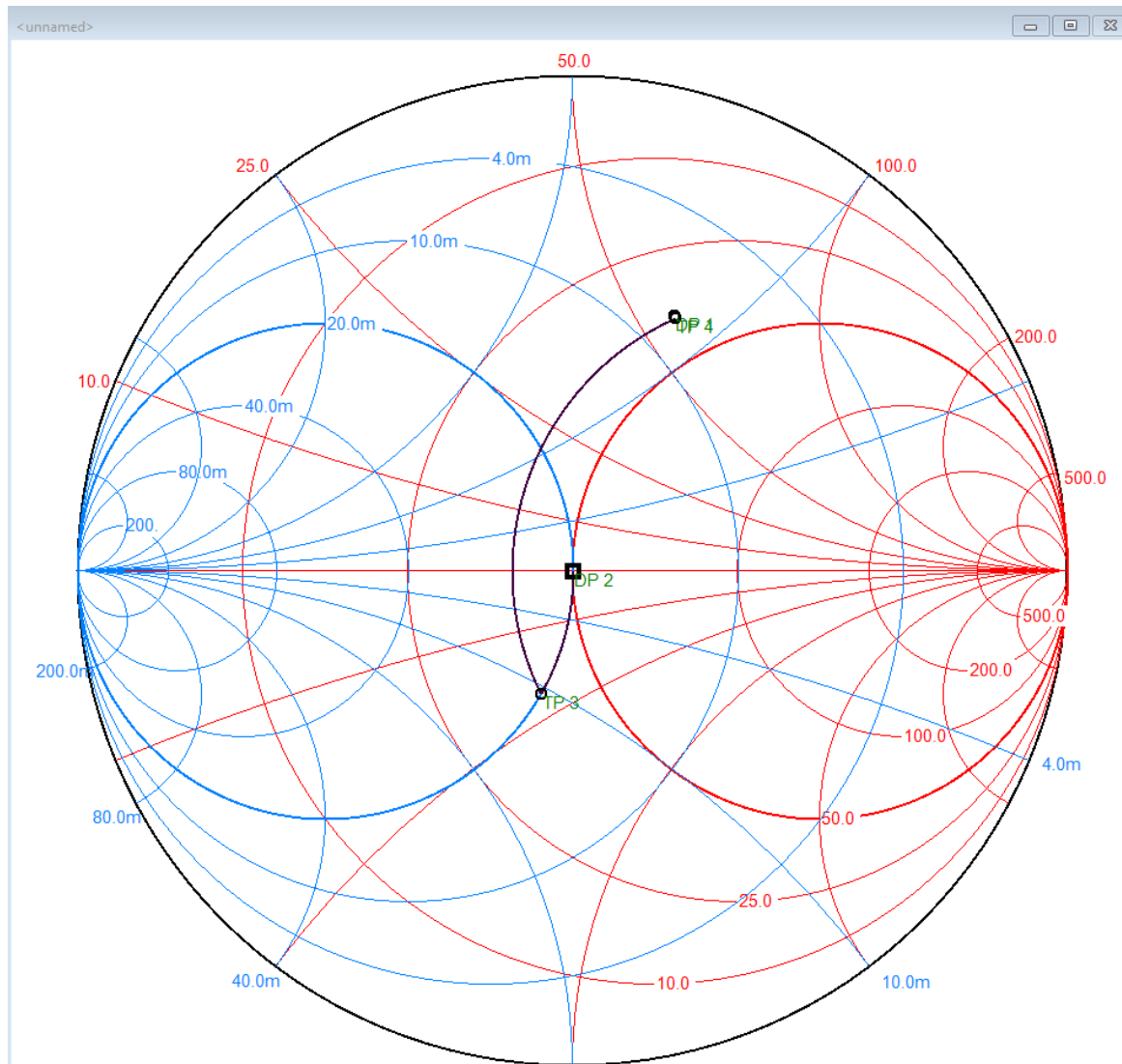
$$2\pi f L = 1.63 * 50$$

$$L = 1.63 * 50 / (2\pi f)$$

$$L = 2.6\text{nH}$$



Matching network: Problem 14 –solution (2/2)



Datapoints				
Start DP	Point	Z	Q	Frequency
☐	DP 1	(38.462 + j57.692) Ω	Q=1.500	5.000GHz
☒	DP 2	(50.000 + j0.000) Ω	Q=0.000	5.000GHz
	TP 3	(38.999 - j20.713) Ω	Q=0.531	5.000GHz
	TP 4	(38.999 + j57.608) Ω	Q=1.477	5.000GHz

Circles		
Visible	Highlighted	Details

Cursor	
Return Loss	0.16 dB
Q	23.78
Y	(3.65-j86.89)mS
Zo	50.0Ω
VSWR	108.95 :1
Γ	0.982 / 154.118 °
Z	(0.48+j11.49)Ω
Freq	5.000GHz